INCH-POUND

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PERFORMANCE SPECIFICATION

COUPLING, AERIAL PRESSURE REFUELING TYPE MA-5

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Comments, suggestions, or questions on this document should be addressed to Air Force Life Cycle Management Center (AFLCMC/EZSS), 2145 Monahan Way, Bldg 28, Wright-Patterson AFB, OH, 45433-7017, or e-mailed to engineering.standards@us.af.mil. Since contact information can change, you should verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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CONTENTS

PARAGRAPH PAGE

1. SCOPE 1
1.1 Scope 1
1.2 Classification 1

2. APPLICABLE DOCUMENTS 1
2.1 General 1
2.2 Government documents 1
2.3 Non-Government publications 3
2.4 Order of precedence 4

3. REQUIREMENTS 4
3.1 Qualification 5
3.2 Government-loaned property 5
3.3 Assembly 5
3.4 Materials 5
3.5 Selection of specifications and standards 6
3.6 Flight/Ground test 7
3.7 Construction 7
3.8 Operation 8

4. VERIFICATION 13
4.1 Classification of inspections 13
4.2 Test and evaluation 15
4.3 Verification requirements 15

5. PACKAGING 27
5.1 Packaging 27
5.2 Marking of shipments 27

6. NOTES 27
6.1 Intended use 28
6.2 Acquisition requirements 28
6.3 Qualification 28
6.4 Definitions 28
6.5 Government-furnished and Government-loaned property 29
6.6 Subject term (key word) listing 29
6.7 International standardization agreement implementation 29
6.8 Changes from previous issue 29

CONCLUDING MATERIAL 48

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CONTENTS

TABLES PAGE

TABLE I. Qualification sequence No. 1 30
TABLE II. Qualification sequence No. 2 31
TABLE III. Fluid resistance and extreme temperature 32
TABLE IV. Contaminant mixture 34
TABLE V. Impact and endurance test 35
TABLE VI. Acceptance sequence 36
TABLE VII. Verification tolerances 36
TABLE VIII. Lot sequence 36

FIGURES

FIGURE 1. Coupling dimensions No. 1 37
FIGURE 2. Coupling dimensions No. 2 38
FIGURE 3. Coupling dimensions No. 3 39
FIGURE 4. Gauge point dimension 40
FIGURE 5. Coupling components 41
FIGURE 6: Ambient-sensing port plug dimensions 42
FIGURE 7. Vibration test curve 43
FIGURE 8. Surge suppression and pressure regulation test setup 44
FIGURE 9. Impact and endurance test setup 45
FIGURE 10. Load application points 46
FIGURE 11. Vibration test setup 47

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1. SCOPE

1.1 Scope. This specification covers aerial refueling fuel reception couplings that provide redundant regulated fuel pressure and suppression of surges generated by the receiver aircraft using the probe and drogue method of aerial refueling.

1.2 Classification. The couplings covered in this document are considered type MA-5 and incorporate two fuel-pressure-regulation mechanisms, which are redundant to each other.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-A-8625	-	Anodic Coatings for Aluminum and Aluminum Alloys
MIL-DTL-5541	-	Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-DTL-5624	-	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-DTL-25988	-	Rubber, Fluorosilicone Elastomer, Oil-and-Fuel Resistant, Sheets, Strips, Molded Parts, and Extruded Shapes
MIL-DTL-27426	-	Rings, Retaining, Spiral (Uniform Cross Section)
MIL-DTL-83133	-	Turbine Fuel, Aviation, Kerosene Type, JP-8 (NATO F-34), NATO F-35, and JP-8+100 (NATO F-37)
MIL-I-8500	-	Interchangeability and Replaceability of Component Parts for Aerospace Vehicles
MIL-N-25161	-	Nozzle, Aerial Pressure Refueling, Type MA-2
MIL-PRF-7024	-	Calibrating Fluids, Aircraft Fuel System Components

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MIL-S-7742	-	Screw Threads, Standard, Optimum Selected Series, General Specification for (Inactive for New Design)

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-129	-	Military Marking for Shipment and Storage
MIL-STD-130	-	Identification Marking of U.S. Military Property
MIL-STD-190	-	Identification Marking of Rubber Products
MIL-STD-464	-	Electromagnetic Environmental Effects Requirements For Systems
MIL-STD-810	-	Environmental Engineering Considerations and Laboratory Tests
MIL-STD-889	-	Dissimilar Metals

(Copies of these documents are available from http://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

MILITARY STANDARDS

MS24354	-	Drogue Cone, Nozzle and Reception Coupling - Type MA-2, Flight-Pressure-Refueling System, Assembly of. (Inactive for New Design)
MS24358	-	Fitting, Hose End - Type MA-2, Flight Pressure Refueling, Reception Coupling, Outline Dimensions for. (Inactive for New Design)
MS24362	-	Sleeve - Type MA-2, Flight Pressure Refueling, Reception Coupling. (Inactive for New Design)
MS24361	-	Ring, Split – Type MA-2, Flight Pressure Refueling, Reception Coupling

(Copies of these documents are available from http://quicksearch.dla.mil.)

TECHNICAL EXHIBITS

ENFE-90-2	-	Tester, Aerial Refueling Coupling, General Exhibit For

(Copies of this document are available from AFLCMC/EZFA, 2145 Monahan Way, Bldg 28,

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Wright-Patterson AFB, OH 45433-7017.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

NASM20995	-	Wire, Safety or Lock
NASM25027	-	Nut, Self-locking, 250 °F, 450 °F, and 800 °F
NASM33540	-	Safety Wiring, Safety Cabling, Cotter Pinning, General Practices for
NASM33588	-	Nut, Self-locking, Aircraft, Reliability And Maintainability Usage Requirements for

(Copies of these documents are available online at http://www.aia-aerospace.org or from the Aerospace Industries Association, 1000 Wilson Boulevard, Suite 1700, Arlington, VA 22209-3928.)

ASTM INTERNATIONAL

ASTM D1655	-	Standard Specification for Aviation Turbine Fuels
ASTM D5900	-	Standard Specification for Physical and Chemical Properties of Industry Reference Materials (IRM)

(Copies of these documents are available online at http://www.astm.org or from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B46.1	-	Surface Texture (Surface Roughness, Waviness and Lay)
ASME Y14.100	-	Engineering Drawing and Related Document Practices

(Copies of these documents are available online at http://www.asme.org or from the American Society of Mechanical Engineers, P.O. Box 2300, Fairfield, NJ 07707-2300.)

GOSUDARSTVENNYE STANDARTY STATE STANDARD (GOST-Russian)

GOST 10227 (TS-1)	-	Jet Fuels Specifications

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(Copies of this document are available from GOST Interstandard, 117049 Moscow, B-29 Russia, Attn: Mr. Nickolay Usov, nusov@gost.ru or in English from www.ihs.com.)

MINISTRY OF DEFENCE

Defence Standard 91-091	-	Turbine Fuel, Aviation Kerosine Type, JET A-1, NATO Code: F-35, Joint Service Designation: AVTUR

(Copies of these documents are available online at https://www.dstan.mod.uk.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE-AMS-STD-595	-	Colors Used in Government Procurement
SAE AMS1424	-	Deicing/Anti-Icing Fluid, Aircraft, SAE Type I
SAE AMS1428	-	Fluid, Aircraft Deicing/Anti-Icing, Non-Newtonian, (Pseudoplastic), SAE Types II, III, and IV
SAE AMS2175	-	Castings, Classification and Inspection of
SAE AMS2447	-	Coating, Thermal Spray, High Velocity Oxygen/Fuel Process
SAE AMS7259	-	Rubber: Fluorocarbon (FKM), High Temperature/Fluid Resistant, Low Compression Set/ 85 to 95 Hardness, For Seals In Fuel Systems and Specific Engine Oil Systems
SAE AS83485	-	O-Ring Molded From AMS7287 Material
SAE ARP868	-	Method-Pressure Drop Tests for Fuel System Components
SAE AS568	-	Aerospace Size Standard for O-rings
SAE AS8879	-	Screw Threads - UNJ Profile, Inch Controlled Radius Root with Increased Minor Diameter
SAE AS29513	-	Packing, Preformed, Hydrocarbon Fuel Resistant, “O” Ring

(Copies of these documents are available online at http://www.sae.org or from the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15090.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

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3.1 Qualification. The couplings furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.3 and 6.3).

3.2 Government-loaned property. The qualifying activity shall supply a KC-130 low-speed drogue basket, National Stock Number (NSN) 1680-00-068-1356, and a KC-130 high-speed drogue basket, NSN 1670-01-422-3223, for qualification purposes.

3.3 Assembly. The coupling shall consist of an assembled unit conforming to the dimensions shown in figure 1, figure 2, figure 3, and figure 4.

NOTE: Attachment parts MS24358, MS2436l, MS24362, SAE AS29513, and a spiral retaining ring conforming to MIL-DTL-27426 are not considered a portion of the coupling. Installation features of these attachments are shown on MS24354. The basic components of a coupling and their typical location are shown in figure 5.

3.4 Materials.

3.4.1 Fluid resistance. All materials used in the construction of the coupling shall be resistant to the following fluids:

a. Jet fuels conforming to MIL-DTL-5624, MIL-DTL-83133, and GOST 10227 (TS-1).

b. Jet-A, Jet-Al, Jet-B, and other commercially available turbine fuels conforming to ASTM D1655 and Def Stan 91-091.

c. NATO fuels F-24, F-34, F-35, F-37, F-40, and F-44.

d. Test fluids conforming to MIL-DTL-5624; ASTM D5900, Industry Reference Materials (IRM) 901 and 903; and MIL-PRF-7024, Type II.

3.4.1.1 Deicing fluid resistance. All materials used in the construction of the coupling shall be resistant to deicing fluid conforming to SAE AMS1424 and SAE AMS1428.

3.4.2 Corrosion-resistant materials. All materials shall be corrosion-resistant or treated to resist corrosion from fuels, salt spray, and atmospheric conditions to be encountered in storage or normal service.

3.4.2.1 Protective coatings. Protective coatings shall not chip, crack, or flake.

3.4.2.2 Aluminum protection. Aluminum-alloy parts shall be anodized in accordance with MIL-A-8625, chemical-conversion coated per MIL-DTL-5541, or treated as approved by the qualifying activity. Flame coating may be used in accordance with SAE AMS2447.

3.4.2.3 Prohibited coatings. The following coatings shall not be allowed as a protective coating:

a. Paint (unless specified in 3.7.10 to 3.7.10.4).

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b. Cadmium.

c. Chromium.

3.4.3 Prohibited materials. Magnesium, chromium, zinc, cadmium, and copper, or their alloys, shall not be used.

3.4.3.1 Impact materials. Polytetrafluoroethylene (PTFE) shall not be used in any location where impact loads may be encountered.

3.4.3.2 Paint. Paint shall not be used, except for identification markings on couplings as specified in 3.7.10 to 3.7.10.4.

3.4.3.3 Age-sensitive materials. Age-sensitive materials shall not be used, except those specified in 3.7.1.

3.4.4 Castings. Castings shall be clean, sound, and free from voids, porosity, cracks, and other defects.

3.4.4.1 Critical castings. Critical (load-bearing) castings shall be in accordance with SAE AMS2175, Class 1, Grade B.

3.4.4.2 Non-critical castings. All castings, other than critical (load-bearing), shall be classified in accordance with SAE AMS2175, Class 1, Grade C.

3.4.5 Dissimilar metals. Dissimilar metals shall not be used in contact with each other, as specified in MIL-STD-889, unless protected against electrolytic corrosion.

3.4.6 Electrical protection.

3.4.6.1 Bonding. The metal parts of the coupling assembly shall be designed to meet the electrical-bonding requirements of MIL-STD-464, paragraph 5.11.3.

3.4.6.2 Lightning protection. The coupling assembly shall be designed to meet the lightning direct effects requirements of MIL-STD-464, paragraph 5.5.

3.4.6.3 Fuel-vapor ignition protection. The coupling shall be designed consistent with the requirements of MIL-STD-464, paragraph 5.11.4, which precludes the possibility of sustained fire or explosion from lightning or electrical faults.

3.4.6.4 Static charging. The coupling assembly fuel-transfer equipment shall be bonded in accordance with MIL-STD-464 for all mating seals/surfaces to prevent buildup of static electrical charge per MIL-STD-464, paragraphs 5.8 and 5.8.1.

3.4.6.5 Total bonding. The total bonding resistance across the coupling assembly shall not exceed the total number of mechanical interfaces in the designed bonding path multiplied by 2.5 milliohms, not to exceed 10 milliohms overall.

3.5 Selection of specifications and standards. Specifications and standards for necessary

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commodities and services not specified herein shall be selected by standard commercial practices or as specified in the contract.

3.6 Flight/Ground test. The contract shall specify the requirements for ground and flight tests for final U.S. Government approval of the coupling.

3.7 Construction. The coupling shall be designed to prevent incorrect assembly; prevent parts from working loose during shipment, storage, or operational use; and withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and use.

3.7.1 O-rings, packing, and seals. O-ring, packing, and seal design and construction shall ensure maximum protection against leakage and damage due to friction, and conform to SAE AS568, MIL-DTL-25988, SAE AS83485, SAE AMS7259, or as approved by the qualifying activity.

3.7.2 Threaded parts. Screw threads shall conform to MIL-S-7742, SAE AS8879, or as approved by the qualifying activity.

3.7.2.1 Prohibited threads. Pipe threads shall not be used.

3.7.2.2 Locking of threaded parts. All threaded parts shall be positively locked by safety wiring, self-locking nuts, or other methods as approved by the qualifying activity.

3.7.2.2.1 Safety wiring. Threaded parts using safety wiring shall conform to NASM20995 and be installed in accordance with NASM 33540.

3.7.2.2.2 Self-locking nuts. Threaded parts using self-locking nuts shall conform to NASM 25027 and be used in accordance with NASM 33588.

3.7.2.2.3 Prohibited locking. The use of lock washers, set screws, cotter pins, or staking shall not be permitted unless specifically approved by the qualifying activity.

3.7.3 Weight. The maximum weight of the coupling without the shroud (see 6.4) shall not be greater than 22 pounds.

3.7.4 Finishes. The coupling finish shall be smooth and free from sharp edges.

3.7.4.1 Surface roughness. Surface roughness, where indicated on figure 1, figure 2, figure 3, and figure 4, shall be as specified in accordance with ASME B46.1.

3.7.4.2 Housing. The external surface of the swivel-ball joint (see 6.4) bell housing (figure 5) shall be gloss white as specified in SAE-AMS-STD-595/17875.

3.7.5 Reliability. The coupling shall be designed and constructed in a manner that will ensure the highest degree of operational reliability under all service conditions when operating with nozzles qualified to MIL-N-25161, Type 2, Class A or B.

3.7.6 Maintainability. The coupling shall be capable of being maintained with

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commercially available tools and equipment by service maintenance personnel having minimum training.

3.7.6.1 Routine maintenance. The coupling shall not require the replacement of parts due to the performance of routine maintenance (see 6.4), to include adjusting the latch settings and performing pressure checks.

3.7.7 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable in accordance with MIL-I-8500.

3.7.8 Dimensions. The coupling dimensions shall be in accordance with figure 1, figure 2, figure 3, and figure 4.

3.7.9 Lubrication. The coupling shall operate without requiring lubrication other than the fluid being transferred.

3.7.10 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130 or MIL-STD-190, as applicable.

3.7.10.1 Application. The identification shall be permanent and not peel off, crack, or loosen with age, environmental conditions, or damage resulting from aerial refueling operations.

3.7.10.2 Location. The location selected for markings shall not affect the performance of the coupling.

3.7.10.3 Markings. The coupling assembly markings shall be of a contrasting color from the background and at a minimum include: part number, serial number, and latch-force setting.

3.7.10.4 Part number. The item identification and part number requirements of ASME Y14.100 shall govern the manufacturer's part numbers and changes thereto.

3.7.11 Workmanship. The coupling shall be uniform in quality, clean, free from defects, parts properly aligned, tightly fitted together, and all burrs removed.

3.7.12 Incorrect assembly. The coupling shall be designed to prevent incorrect assembly.

3.8 Operation. The coupling shall conform to figure 1, figure 2, figure 3, and figure 4 and be compatible with aerial refueling nozzles conforming to MIL-N-25161 to accomplish fuel transfer in flight. The coupling shall provide surge pressure protection (receiver generated surge) for the receiver aircraft along with steady-state pressure control to prevent excessive tanker pressures from entering the receiver during refueling. This shall be accomplished by means of two pressure-regulating mechanisms, redundant to each other and integral to the coupling. The failure of either mechanism shall not prevent the coupling from providing surge control and pressure regulation as specified. Upon engagement with the nozzle, the coupling master seal shall make contact with the nozzle's sleeve-lip seal surface prior to the nozzle's sleeve starting to open. Upon disengagement, fuel flow shall be automatically shut off in both the coupling and the nozzle prior to separation between the coupling master seal and the nozzle sleeve's sealing lip at disconnect angles up to 22.5 degrees off center with no binding or damage incurred. This

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operation shall be accomplished without the use of any external source of power other than the forces generated by the receiver aircraft to separate the nozzle and coupling.

3.8.1 Regulation.

3.8.1.1 Regulation capability. Unless otherwise specified in the contract, the coupling shall regulate the outlet pressure, whether one or both regulators are active, to 60 psig maximum at flow rates from 10 cc/min to 1 gallon per minute (gal/min) and 50±5 psig at flow rates of 1 to 600 gal/min with an inlet pressure up to 120 psig.

3.8.1.2 Leakage flow regulation. With a coupling inlet fluid pressure of 60, 80, 100, and 120 psig, the regulated coupling outlet pressure shall not be more than 60 psig plus 10 percent of the inlet pressure as the flow rate is decreased from 10 cc/min to 10 drops per minute or less, whether one or both regulators are active.

3.8.1.2.1 High inlet pressure regulation. With a coupling inlet fluid pressure greater than 120 psig and up to 240 psig, and a flow rate of 10 drops per minute or less, the regulated coupling outlet pressure shall not be more than 120 psig.

3.8.1.3 Ground support equipment. The coupling shall be compatible with current ground support equipment, unless agreed upon by the qualifying activity. To accomplish one of several functions, current ground support equipment (NSN 4920-01-186-5820/4920-01-483-2490 and reference ENFE-90-2) allows for ambient-pressure-sensing ports with a circumference spacing of 120 degrees for 3-port models and 90 degrees for 4-port models, for each pressure regulation device. Ground support equipment ambient-sensing port plugs are shown in figure 6.

3.8.1.4 Independent regulator test. The coupling shall be designed such that each regulator’s operation may be independently verified.

3.8.1.4.1 Moisture. Moisture accumulation shall not affect coupling functionality.

3.8.1.5 Pressure-sensing port. The coupling shall be equipped with a fuel-pressure-sensing port downstream from the internal regulation mechanisms.

3.8.1.6 Pressure relief. Regulator designs, which allow trapping of pressures within the coupling that could hamper engagement, shall include an internal pressure-relief system to dissipate the trapped pressure to comply with the engagement-force requirements of 3.8.3.1 within 3 seconds.

3.8.1.7 Surge suppression. The coupling shall be capable of suppressing surges generated by the receiver, whether one or both regulators are active, such that peak pressures downstream of the coupling are not greater than 120 psig and no damage is sustained by the coupling.

3.8.2 Alignment.

3.8.2.1 Deflection. The coupling shall incorporate a swivel-ball joint having a range of deflection within a 45-degree or greater angle cone, the centerline of which coincides with the

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axial centerline of the coupling.

3.8.2.1.1 Deflection torque. The deflection torque required to move the coupling ball (with fuel-wetted seals) throughout the complete angular range of deflection shall not be less than 98 inch-pounds (in-lbs) or greater than 216 in-lbs.

3.8.2.2 Axial rotation. The coupling swivel-ball joint shall facilitate alignment with a MIL-N-25161 compliant MA-2 nozzle and permit 360 degrees of axial rotation about the axial centerline of the coupling body when engaged with an MA-2 nozzle.

3.8.2.2.1 Axial-rotation torque. The torque required to rotate the coupling body about the swivel-ball joint and coupling centerline shall not be less than 8 foot-pounds (ft-lbs) or greater than 50 ft-lbs with the coupling filled with fuel.

3.8.3 Latching mechanism. The coupling shall have a minimum of three latching mechanisms.

3.8.3.1 Engagement. The force required to engage the coupling with an MA-2 receiver aerial refueling probe nozzle having a 50-pound sleeve force shall not be greater than 155 pounds with a 10 psig minimum static fuel pressure in the coupling.

3.8.3.2 Disengagement. The latching mechanisms shall be such that the force required to disengage the nozzle from the coupling is adjustable between 300 and 550 pounds with 0 psig fuel pressure in the coupling, and without change or addition of components.

NOTE 1: The term “latch-force setting” refers to the overall force required to disengage a probe nozzle from the coupling on centerline with 0 psig fuel pressure, also referred to as the “disengagement force.”

NOTE 2: Unless otherwise specified in the contract, the latch-force setting shall be 420±20 pounds with 0 psig fuel pressure and 620±45 pounds with 50 psig fuel pressure.

3.8.3.2.1 Fuel-pressure force. The force required to disengage the nozzle shall increase at the rate of 4±0.5 pounds for each 1.0 psig increase in pressure downstream of the regulator at any fixed latch-force setting.

3.8.3.2.2 Surge disengagement. Surges up to 120 psig shall not result in nozzle disengagements.

3.8.3.2.3 Maximum disengagement force. The force to separate the nozzle from the coupling (with 50 psig of fuel pressure) with a latch-force setting of 520±20 lbs shall not exceed 830 lbs when the disconnect angle of the swivel-ball joint is at the 22.5 degree position.

3.8.4 Proof pressure.

3.8.4.1 Engaged proof pressure. The coupling, while engaged with a nozzle, shall withstand a minimum internal hydrostatic pressure (see 6.4) of 120 psig with no external leakage (see 6.4), permanent deformation, structural failure, or other failure which would render it

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inoperable.

3.8.4.2 Disengaged proof pressure. The coupling, while not engaged with a nozzle, shall withstand a minimum internal hydrostatic pressure (see 6.4) of 240 psig with no external leakage (see 6.4), permanent deformation, structural failure, or other failure which would render it inoperable.

3.8.5 Ultimate (burst) pressure.

3.8.5.1 Engaged ultimate pressure. The coupling, while engaged with a nozzle, shall withstand a minimum internal hydrostatic pressure (see 6.4) of 360 psig with no external leakage (see 6.4), or structural failure.

3.8.5.2 Disengaged ultimate pressure. The coupling, while not engaged with a nozzle, shall withstand a minimum internal hydrostatic pressure (see 6.4) of 360 psig with no external leakage (see 6.4), or structural failure.

3.8.6 Leakage and spillage.

3.8.6.1 Initial allowances. The initial allowable leakage and spillage shall be as follows:

a. When the engaged coupling and nozzle are subjected to constant inlet pressures of 5 and 120 psig, there shall be no external leakage.

b. When the coupling alone is subjected to constant inlet pressures of 5 and 120 psig, there shall be no external leakage.

c. When the engaged coupling and nozzle are subjected to an inlet fluid pressure of 120 psig and a fluid flow of 100 to 600 gal/min, the spillage shall not be greater than 100 cc per disengagement.

d. When the coupling is engaged and disengaged on a nozzle with an inlet fluid pressure of 10 psig, spillage shall not be greater than 25 cc per engagement or disengagement.

3.8.6.2 Subsequent allowances. The allowable leakage and spillage subsequent to significant coupling operations (as defined by the completion of the test sequence prior to the first “Leakage and Spillage (Subsequent)” test call-out in TABLE I and TABLE II) shall be as follows:

a. When the engaged coupling and nozzle are subjected to constant inlet pressures of 5 and 120 psig, there shall be no seepage in excess of one drop per minute.

b. When the coupling alone is subjected to constant inlet pressures of 5 and 120 psig, there shall be no seepage in excess of one drop per minute.

c. When the engaged coupling and nozzle are subjected to an inlet fluid pressure of 120 psig and a fluid flow of 100 to 600 gal/min, the spillage shall not be greater than 100 cc per disengagement.

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d. When the coupling is engaged and disengaged on a nozzle with an inlet fluid pressure of 10 psig, spillage shall not be greater than 25 cc per engagement or disengagement.

3.8.7 Pressure drop. The total pressure drop through the coupling engaged with an Eaton, MA-2 nozzle (part number 60318-2B), at 60 to 90 °F, using Grade JP-5 (or equivalent), and with regulators inactive shall not be greater than 18 psid at 600 gal/min.

3.8.8 Extreme temperature. The coupling shall meet its functional requirements after exposure to extreme temperatures as specified in TABLE III.

3.8.9 Contaminated fuel. The coupling shall meet its functional requirements after exposure to contaminants specified in TABLE IV.

3.8.10 Design limit loads. The coupling shall be subjected to the following minimum loads with no permanent deformation or other failure which would render it inoperable.

a. A 1,500-pound tension load in combination with a 1,500-pound radial load.

b. A 2,500-pound compression load with a 1,500-pound radial load.

c. A 1,500-pound tension load.

d. A 2,500-pound compression load.

e. A 1,500-pound radial load.

3.8.11 Nozzle-coupling compatibility. The coupling shall be compatible with all nozzles on the current Qualified Products List (QPL), QPL-25161, such that it engages and disengages the nozzle with no hang-ups (including rotation about the nozzle) or excess force, and meets the requirements of 3.8.3.1 and 3.8.3.2 to 3.8.3.2.3.

3.8.12 Altitude. When the ambient coupling pressure is reduced to 4.17 psia, the coupling's regulated outlet pressure shall not decrease more than 15 psig, and there shall be no external leakage.

3.8.13 Environmental requirements.

3.8.13.1 Salt fog. The coupling shall meet its performance requirements after exposure to salt fog.

3.8.13.2 Sand and dust. The coupling shall meet its performance requirements after exposure to both sand and dust.

3.8.13.3 Vibration. The coupling shall meet its performance requirements after being exposed to the vibration requirements illustrated in figure 7.

3.8.13.4 Impact and endurance.

3.8.13.4.1 Impact. When the coupling is dropped as specified in TABLE V for impact,

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leakage shall not be greater than 25 cc per in-line engagement or disengagement.

3.8.13.4.2 Endurance. When the coupling is dropped as specified in TABLE V for endurance, leakage shall not be greater than 25 cc per in-line engagement or disengagement and 100 cc per angular disengagement.

3.8.14 Disassembly and inspection. Parts shall not show any distortion, weakness, or abnormal wear (see 6.4) after exposure to testing outlined in section 4.1.1.

3.8.15 Functional.

3.8.15.1 Deflection. The coupling shall incorporate a swivel-ball joint having a range of deflection of ±22.5 degrees minimum from the coupling centerline.

3.8.15.2 Disengaged leakage. When the coupling alone is subjected to constant inlet pressures of 5 and 60 psig, there shall be no external leakage (initial) and no seepage in excess of one drop per minute (subsequent).

3.8.15.3 Engaged leakage. When the engaged coupling and nozzle are subjected to constant inlet pressures of 5 and 60 psig, there shall be no external leakage (initial) and no seepage in excess of one drop per minute (subsequent).

3.8.15.4 Pressure relief. Regulator designs, which allow trapping of pressures within the coupling that could hamper engagement, shall include an internal pressure-relief system to dissipate the trapped pressure to comply with the engagement force requirements of 3.8.3.1 within 3 seconds.

3.8.15.5 Regulators. Unless otherwise specified in the contract, the coupling shall regulate the outlet pressure, whether one or both regulators are active, to 60 psig maximum at flow rates from 10 cc/min to 1 gal/min and 50±5 psig at flow rates of 1 to 600 gal/min with an inlet pressure up to 120 psig. With a coupling inlet fluid pressure of 60, 80, 100, and 120 psig, the regulated coupling outlet pressure shall not be more than 60 psig plus 10 percent of the inlet pressure as the flow rate is decreased from 10 cc/min to 10 drops per minute or less, whether one or both regulators are active.

3.8.15.6 Engagement and disengagement. The force required to engage couplings of all types with an MA-2 receiver aerial refueling probe nozzle having a 50-pound sleeve force shall not be greater than 155 pounds with a 10 psig minimum static fuel pressure in the coupling. The latching mechanisms shall be such that the force required to disengage the nozzle from the coupling is adjustable between 300 and 550 pounds with 0 psig fuel pressure in the coupling, and without change or addition of components. The force required to disengage the nozzle shall increase at the rate of 4±0.5 pounds for each 1.0 psig increase in pressure downstream of the regulator at any fixed latch-force setting.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

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a. Qualification inspection (see 4.1.1).

b. Conformance inspection (see 4.1.2).

4.1.1 Qualification inspections/tests. Qualification tests shall use at least two couplings and consist of the complete series of tests and inspections specified below, performed in the order listed. If a different test order is desired, it must be approved by the qualifying activity prior to test start.

a. Inspection/Test sequence No. 1 – A single qualification coupling shall be used for the sequence of inspections/tests in TABLE I.

b. Inspection/Test sequence No. 2 – Up to two qualification couplings can be used for the sequence of tests in TABLE II. The column denoted by “1 Unit” shows the tests that shall be accomplished if a single unit (“A”) is used, and the column denoted by “2 Units” shows the tests that shall be accomplished for each coupling (“A” and “B”).

4.1.1.1 Qualification failures. If the part fails during qualification testing, the test shall be terminated and the qualifying activity notified. The qualifying activity shall determine the retest requirements.

4.1.1.2 Samples for government testing. Test couplings for Government or procuring agency testing shall be submitted to the qualifying activities as specified in the contract and consist of two couplings identified with the manufacturer’s own part number.

4.1.1.3 Government tests. At the option of the qualifying activity, the test couplings that were submitted to the Government (see 4.1.1.2) shall be subjected to flight, service, ground, and laboratory tests in accordance with the requirements of the qualifying activity.

4.1.2 Acceptance.

4.1.2.1 Acceptance inspections/tests. Each coupling shall be subjected to the inspections and tests in TABLE VI.

4.1.2.2 Lots. A lot shall consist of couplings of the same part number essentially manufactured under the same conditions and submitted for inspection at the same time.

4.1.2.3 Acceptance failures.

4.1.2.3.1 Rejection. When one or more couplings from a lot fail to meet the requirements of this specification, acceptance of all couplings in the lot shall be withheld until the extent and cause of the failure are determined.

4.1.2.3.2 Resubmittal. Before resubmitting, the full particulars concerning the failures and necessary corrective actions shall be furnished to the qualifying activity.

4.1.2.3.3 Retest. After corrective actions have been taken, all conformance inspections shall be repeated.

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4.1.2.3.4 Resubmittal prohibited. Couplings rejected after retest shall not be submitted for further testing without specific approval of the qualifying activity.

4.2 Test and evaluation.

4.2.1 Test conditions. Unless otherwise specified, all tests shall be performed at atmospheric pressure and ambient air temperature between 60 and 90 °F with the test fluid conforming to MIL-DTL-5624, Grade JP-5, or MIL-PRF-7024.

4.2.2 Tolerances. Unless otherwise specified, test pressures and flow rates within the verification section shall have tolerances in accordance with TABLE VII.

4.2.3 Test nozzles. Unless otherwise specified, nozzles used in the following tests and evaluations shall be any nozzle on the QPL for MIL-N-25161.

4.2.4 Test cycles. Unless otherwise specified, engagement of the nozzle, nozzle sealing in the coupling, and the complete disengagement of the nozzle from the coupling shall constitute one complete cycle of operation.

4.3 Verification requirements.

4.3.1 Qualification. Not applicable.

4.3.2 Government-loaned property. Not applicable.

4.3.3 Assembly. The coupling shall be inspected for conformance to the dimensions shown in figure 1, figure 2, figure 3, and figure 4.

4.3.4 Materials.

4.3.4.1 Fluid resistance. All materials used in the construction of the coupling shall be shown to be resistant to the fluids listed in 3.4.1 and by testing in accordance with TABLE III.

4.3.4.1.1 Deicing fluid resistance. Testing shall be performed in accordance with MIL-STD-810, Method 504.2, Procedure I, with “occasional” length of exposure using deicing fluids listed in 3.4.1.1.

4.3.4.1.1.1 Follow-on tests. Immediately after the test in 4.3.4.1.1, the coupling shall pass the leakage test of 4.3.8.6.2 (subsequent) and the functional tests of 4.3.8.15 to 4.3.8.15.6.

4.3.4.2 Corrosion-resistant materials. The coupling drawings shall be inspected for corrosion-resistant protective measures in compliance with 3.4.2.

4.3.4.2.1 Protective coatings. After qualification testing, the coupling shall be inspected for chipping or flaking of protective coatings.

4.3.4.2.2 Aluminum protection. The coupling drawings shall be inspected to confirm aluminum-alloy parts are protected in accordance with 3.4.2.2.

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4.3.4.2.3 Prohibited coatings. The coupling drawings and hardware shall be inspected for exclusion of the protective coatings given in 3.4.2.3.

4.3.4.3 Prohibited materials. The coupling materials list shall be inspected for exclusion of prohibited materials given in 3.4.3.

4.3.4.3.1 Impact materials. Analysis and inspection of coupling drawings shall show that PTFE is not used in any location where impact loads may be encountered.

4.3.4.3.2 Paint. The coupling drawings and hardware shall be inspected for exclusion of paint, except as specified in 3.7.10 to 3.7.10.4.

4.3.4.3.3 Age-sensitive materials. The coupling drawings shall be inspected for exclusion of age-sensitive materials, except those specified in 3.7.1.

4.3.4.4 Castings. The coupling castings shall be inspected for cleanliness, soundness, and the lack of voids, porosity, cracks, and other defects.

4.3.4.4.1 Critical castings. The coupling drawings shall be inspected for critical castings compliance to 3.4.4.1.

4.3.4.4.2 Non-critical castings. The coupling drawings shall be inspected for castings, other than critical (load-bearing), compliance to 3.4.4.2.

4.3.4.5 Dissimilar metals. The coupling drawings shall be inspected for dissimilar metals compliance to 3.4.5.

4.3.4.6 Electrical protection.

4.3.4.6.1 Bonding. The coupling drawings shall be inspected to ensure the materials and surface coatings for all faying surfaces are appropriate and electrically conductive.

4.3.4.6.2 Lightning protection. Verified by 4.3.4.6.1 and 4.3.4.6.5.

4.3.4.6.3 Fuel-vapor ignition protection. Verified by 4.3.4.6.1 and 4.3.4.6.5.

4.3.4.6.4 Static charging. Verified by 4.3.4.6.1 and 4.3.4.6.5.

4.3.4.6.5 Total bonding. The total coupling resistance shall be measured, with no loads applied, via an electrical-resistance measurement using a milliohm meter capable of reading down to 1 milliohm or less to show it is in accordance with 3.4.6.5.

4.3.5 Selection of specifications and standards. Verification for necessary commodities and services not specified herein shall be in accordance with applicable commercial practices or as specified in the contract.

4.3.6 Flight/Ground test. Verification by ground and flight test shall be as specified in the contract.

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4.3.7 Construction. Not applicable.

4.3.7.1 O-rings, packings, and seals. The coupling drawings, O-rings, packings, and seals shall be inspected for compliance to 3.7.1.

4.3.7.2 Threaded parts. The coupling drawings and hardware shall be inspected for compliance to 3.7.2.

4.3.7.2.1 Prohibited threads. The coupling drawings shall be inspected for compliance to 3.7.2.1.

4.3.7.2.2 Locking of threaded parts. The coupling drawings and hardware shall be inspected for conformance to approved positively-locking methods in accordance with 3.7.2.2.

4.3.7.2.2.1 Safety wiring. For threaded parts using safety wiring, the coupling drawings and hardware shall be inspected for compliance to 3.7.2.2.1.

4.3.7.2.2.2 Self-locking nuts. For threaded parts using self-locking nuts, the coupling drawings and hardware shall be inspected for compliance to 3.7.2.2.2.

4.3.7.2.2.3 Prohibited locking. The coupling drawings and hardware shall be inspected for exclusion of locking methods identified in 3.7.2.2.3.

4.3.7.3 Weight. The coupling, without the shroud, shall be weighed to demonstrate a total weight in compliance with 3.7.3.

4.3.7.4 Finishes. The coupling finish shall be inspected for compliance to 3.7.4.

4.3.7.4.1 Surface roughness. The coupling finish shall be inspected for compliance to 3.7.4.1.

4.3.7.4.2 Housing. The coupling and drawings shall be inspected for an external housing coating in accordance 3.7.4.2.

4.3.7.5 Reliability. This requirement is verified by successful completion of the environmental verifications in 4.3.8.13 to 4.3.8.13.4.2.1.

4.3.7.6 Maintainability. The coupling and maintenance manuals shall be inspected to verify coupling maintenance is in accordance with 3.7.6.

4.3.7.6.1 Routine maintenance. The coupling and maintenance manuals shall be inspected to verify coupling maintenance is in accordance with 3.7.6.1.

4.3.7.7 Interchangeability. The maintenance manuals shall be inspected to verify part interchangeability in accordance with 3.7.7.

4.3.7.8 Dimensions. The coupling hardware shall be measured and drawings inspected for conformance to the dimensions given in figure 1, figure 2, figure 3, and figure 4.

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4.3.7.9 Lubrication. The coupling and drawings shall be inspected to verify compliance to 3.7.9.

4.3.7.10 Identification of product. The coupling equipment, assemblies, and parts shall be inspected for compliance to 3.7.10.

4.3.7.10.1 Application. The coupling equipment, assemblies, and parts identification markings shall be inspected at the completion of qualification testing to ensure there is no peeling, cracking, or loosening of the marking.

4.3.7.10.2 Location. The coupling and drawings shall be inspected to verify the markings do not affect the coupling performance.

4.3.7.10.3 Markings. The coupling and drawings shall be inspected for the coloring and inclusion of the information identified in 3.7.10.3.

4.3.7.10.4 Part number. The coupling and drawings shall be inspected to verify the part numbers conform to ASME Y14.100 identification and part number requirements.

4.3.7.11 Workmanship. The coupling shall be inspected for uniformity, cleanliness, lack of defects, proper alignment, and removal of burrs.

4.3.7.12 Incorrect assembly. The coupling drawings shall be inspected for potential of incorrect assembly.

4.3.8 Operation. Not applicable.

4.3.8.1 Regulation. The following tests shall be conducted with the coupling engaged with an MA-2 nozzle (Eaton, part number 60318-2B), using an apparatus similar to that shown in figure 8.

4.3.8.1.1 Regulation capability. The test shall be conducted with the attitude of the coupling being horizontal +0/-5 degrees and performed as follows for each regulator independently, as well as both concurrently:

a. For fuel-flow rates between and including 10±0.5 cc/min and 1 -0.05/+0 gal/min at coupling inlet fuel pressures of 60, 80, 100, and 120±2 psig, sufficient data (as agreed by the qualifying activity) shall be recorded to plot the regulating characteristics and show regulated downstream pressure does not exceed 60 psig.

b. For fuel-flow rates between and including 1 -0/+0.05 gal/min and 600 -0/+30 gal/min at coupling inlet fuel pressures of 60, 80, 100, and 120±2 psig, sufficient data (as agreed by the qualifying activity) shall be recorded to plot the regulating characteristics and show regulated downstream pressure is 50±5 psig. At the 600-gal/min test condition, the inlet pressure of 120±2 psig is not required to be tested.

c. While maintaining a flow rate of 50 gal/min, the inlet pressure shall be cycled from 60 to 100±2 psig and back to 60 psig for a total of 400 cycles at a rate of six cycles per minute.

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Then, while maintaining a flow rate of 300 gal/min, the inlet pressure shall be cycled from 60 to 100±2 psig and back to 60 psig for a total of 800 cycles at a rate of six cycles per minute. Lastly, while maintaining a flow rate of 600 gal/min, the inlet pressure shall be cycled from 60 to 90±2 psig and back to 60 psig for a total of 800 cycles at a rate of six cycles per minute. The outlet pressure shall not exceed 55 psig, and there shall be no external leakage.

4.3.8.1.1.1 Follow-on tests. Immediately after the test in 4.3.8.1.1(c), the coupling shall pass the functional test of 4.3.8.15 to 4.3.8.15.6.

4.3.8.1.2 Leakage flow regulation. The test shall be performed for each regulator independently, as well as both concurrently. With a coupling inlet fluid pressure of 60±2 psig, the coupling outlet pressure shall be monitored as the flow rate is decreased from 10 cc/min to 10 drops per minute. The regulated outlet pressure shall not be more than 60 psig plus 10 percent of the inlet pressure. The test shall be repeated at inlet pressures of 80, 100, and 120±2 psig.

4.3.8.1.2.1 High inlet pressure regulation. The test shall be performed for each regulator independently, as well as both concurrently. With a coupling inlet fluid pressure of 240±2 psig, the coupling outlet pressure shall be monitored as the flow rate is decreased from 10 cc/min to 10 drops per minute. The regulated outlet pressure shall not be more than 120 psig.

4.3.8.1.3 Ground support equipment. The coupling shall be demonstrated to be compatible with current ground support equipment in accordance with 3.8.1.3.

4.3.8.1.4 Independent regulator test. The coupling shall demonstrate the ability to disable each regulator independently in order to verify proper operation of the remaining functioning regulator.

4.3.8.1.4.1 Moisture. Testing shall be performed in accordance with MIL-STD-810, Method 524.1, Procedure III, with the coupling orientation such that the outlet is 15±5 degrees down from horizontal in the axial direction and the inlet sealed closed. The cold chamber temperature in the procedure shall be 14 °F maximum in step 2. The regulation-capability test in 4.3.8.1.1(a) shall be performed as the “operational check” in step 3 of the procedure after the final cold temperature exposure of step 2. The regulation-capability test shall be performed with a fuel temperature of 14 °F maximum, fuel-flow rate of 30±1.5 cc/min, and fuel inlet pressure of 80±2 psig.

4.3.8.1.4.1.1 Follow-on tests. During the test in 4.3.8.1.4.1 where operational tests are annotated, the coupling shall pass the functional tests of 4.3.8.15 to 4.3.8.15.6 and the leakage tests of 4.3.8.6.2 (subsequent).

4.3.8.1.5 Pressure-sensing port. The coupling and drawings shall be inspected for the inclusion of a fuel pressure-sensing port in accordance with 3.8.1.5.

4.3.8.1.6 Pressure relief. The coupling (outlet and inlet) alone shall be pressurized to 120±2 psig, then the pressure reduced at the inlet to 10±1 psig. The time for the pressure to decay within the coupling to the level where the engagement force meets the requirement of 3.8.3.1 shall be recorded. The recorded time shall be in accordance with 3.8.1.6.

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4.3.8.1.7 Surge suppression. This test shall be conducted using an apparatus similar to that shown in figure 8. With a nozzle inlet fluid pressure of 55±5 psig and fuel line velocities from 5 to 40 feet/second (ft/s) (in increments of 5 ft/s), surge pressures shall be generated by closure of the controlled rate of closure (CRC) valve (equivalent 3-inch, full-flow, ball valve with a constant closure rate within 0.2 seconds from full open to full close). Surge pressures shall be recorded within 6 inches of the coupling inlet and downstream of the coupling within 30 inches of the MA-2 nozzle interface to the probe. Pressure peaks recorded upstream of the coupling are for the test report only, and reflected surge pressure peaks recorded downstream of the coupling shall not be greater than the requirements specified in 3.8.1.7. The test shall be conducted using pressure transducers with a minimum of 1,000-Hz sampling rate and refueling hose inner diameters of 1.5 inches and 2.625 inches. The test shall be repeated five times for each refueling hose configuration. The 2.625-inch diameter hose is not required to accomplish the 40 ft/s test condition. This test must also be performed in the following three ways:

a. With both regulators working.

b. With No. 1 regulator inactive (held in full-open position).

c. With No. 2 regulator inactive (held in full-open position).

4.3.8.2 Alignment.

4.3.8.2.1 Deflection. With either the coupling inlet or outlet flange fixed, the free end of the coupling shall be deflected about the swivel-ball joint to demonstrate the cone of deflection described in 3.8.2.1.

4.3.8.2.1.1 Deflection torque. The coupling shall be filled with test fluid and then drained. The torque required to deflect the coupling about the swivel-ball joint shall then be measured and meet the requirements described in 3.8.2.1.1.

4.3.8.2.2 Axial rotation. With either the coupling inlet or outlet flange fixed, the free end shall be rotated about the swivel-ball joint to demonstrate the axial-rotation requirements described in 3.8.2.2.

4.3.8.2.2.1 Axial-rotation torque. The coupling shall be filled with test fluid and then drained. With either the coupling inlet or outlet flange fixed, the free end shall then be rotated about the swivel-ball joint and the required axial-rotation torque measured. The torque required to rotate the coupling body 360 degrees around the swivel-ball joint about the coupling centerline shall meet the requirements of 3.8.2.2.1.

4.3.8.3 Latching mechanism. The coupling and drawings shall be inspected for the inclusion of at least three latching mechanisms.

4.3.8.3.1 Engagement. Using a test stand similar to that shown in figure 9, the test coupling shall be filled with fuel and pressurized to 10±1 psig. The engaging force shall be as specified in 3.8.3.1. Tapping of the mast to effect engagement shall not be allowed. The nozzle and coupling centerlines shall not be displaced for this test.

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4.3.8.3.2 Disengagement. The coupling shall be engaged with a nozzle and the latching mechanism adjusted to produce a disengagement force of 550 -0/+20 pounds with 0 psig fuel pressure. The coupling shall then be disengaged from the nozzle and the disengagement force recorded. The test shall be repeated with a latch-force setting of 300 -20/+0 pounds.

4.3.8.3.2.1 Fuel-pressure force. The coupling shall be engaged with a nozzle and the latching mechanism adjusted to produce a disengagement force of 550±20 pounds. With the coupling outlet pressure increased to 50±2 psig (no flow), the disengaging force shall increase as specified in 3.8.3.2.1 and the values recorded. The test shall be repeated with a latch-force setting of 320±20 pounds. Then, with a coupling outlet pressure of 50±2 psig and a flow rate of 600±25 gal/min, the disengaging force shall be rechecked. The force shall change no greater than 25 pounds from that obtained in the no-flow test and shall meet the requirements of 3.8.3.2.1.

4.3.8.3.2.2 Surge disengagement. The coupling shall be engaged with a nozzle (using an apparatus similar to that shown in figure 8) and the latching mechanism adjusted to produce a disengagement force of 300±20 pounds. With a coupling inlet fluid pressure of 55±5 psig and flow rate of 600 gal/min, surge pressures shall be generated by closure of the CRC valve (equivalent 3-inch, full-flow, ball valve with a constant closure rate within 0.2 seconds from full open to full close). The coupling shall not disengage from the nozzle.

4.3.8.3.2.3 Maximum disengagement force. The coupling shall be engaged with a nozzle (using an apparatus similar to that shown in figure 8) and the latching mechanism adjusted to produce a disengagement force of 520±45 pounds. With the coupling outlet pressure increased to 50±2 psig (no flow) and the nozzle at a 22.5-degree angle from the centerline of the coupling, the nozzle shall be disengaged and the force measured. The disengaging force shall not exceed 830 pounds in accordance with 3.8.3.2.3.

4.3.8.3.2.4 Follow-on tests. Immediately after the tests in 4.3.8.3.2 to 4.3.8.3.2.3, the coupling shall pass the leakage test of 4.3.8.6.2 (subsequent) and the functional tests of 4.3.8.15 to 4.3.8.15.6.

4.3.8.4 Proof pressure.

4.3.8.4.1 Engaged proof pressure. With the coupling engaged with a nozzle and the nozzle outlet blanked off, the coupling shall be subjected to a hydrostatic pressure of 120±2 psig for a minimum of 5 minutes. There shall be no evidence of external leakage, distortion, or other damage to any part of the coupling throughout the required test period.

4.3.8.4.2 Disengaged proof pressure. The coupling alone shall be subjected to a hydrostatic pressure of 240±2 psig for a minimum of 5 minutes. There shall be no evidence of external leakage, distortion, or other damage to any part of the coupling throughout the required test period.

4.3.8.4.2.1 Follow-on tests. Immediately after the tests in 4.3.8.4.1 and 4.3.8.4.2, the coupling shall pass the functional tests of 4.3.8.15 to 4.3.8.15.6 and the leakage tests of 4.3.8.6.1 (initial).

4.3.8.5 Ultimate (burst) pressure.

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4.3.8.5.1 Engaged ultimate pressure. With the coupling engaged with a nozzle and the nozzle outlet blanked off, the coupling shall be subjected to an internal hydrostatic pressure of 360±2 psig for a minimum of 5 minutes. It shall meet the requirements of 3.8.5.1.

4.3.8.5.2 Disengaged ultimate pressure. The coupling alone shall be subjected to an internal hydrostatic pressure of 360±2 psig for a minimum of 5 minutes. It shall meet the requirements of 3.8.5.2. The pressure shall then be lowered to 60 psig. There shall be no evidence of external leakage.

4.3.8.5.2.1 Follow-on tests. Immediately after the tests in 4.3.8.5.1 and 4.3.8.5.2, the coupling shall pass the leakage tests of 4.3.8.6.2 (subsequent).

4.3.8.6 Leakage and spillage. The attitude of the coupling shall be most conducive for the drainage of leakage which may be retained internally by the coupling for the following tests.

4.3.8.6.1 Initial allowances. This test shall be performed as follows:

a. The engaged coupling and nozzle shall be subjected to constant inlet pressures of 5±0.5 psig and 120±2 psig for a minimum of 6 minutes each. There shall be no external leakage at each pressure in excess of the limit specified in 3.8.6.1(a).

b. The coupling alone shall be subjected to constant inlet pressures of 5±0.5 psig and 120±2 psig for a minimum of 6 minutes each. There shall be no external leakage at each pressure in excess of the limit specified in 3.8.6.1(b).

c. The engaged coupling and nozzle shall be subjected to an inlet fluid pressure of 120±2 psig and a fluid flow of 100 -5/+0 to 600 -0/+30 gal/min in increments of 100 gal/min. The nozzle shall be disengaged at every increment. The spillage per disengagement shall not be greater than the limit specified in 3.8.6.1(c).

d. The coupling shall be subjected to an inlet fluid pressure of 10±1 psig providing unrestricted fuel flow. The nozzle shall be engaged and disengaged a minimum of five times. Spillage per engagement or disengagement shall not be greater than the limit specified in 3.8.6.1(d).

4.3.8.6.2 Subsequent allowances. This test shall be performed as follows:

a. The engaged coupling and nozzle shall be subjected to constant inlet pressures of 5±0.5 psig and 120±2 psig for a minimum of 6 minutes each. There shall be no seepage at each pressure in excess of the limit specified in 3.8.6.2(a).

b. The engaged coupling and nozzle shall be subjected to constant inlet pressures of 5±0.5 psig and 120±2 psig for a minimum of 6 minutes each. There shall be no seepage at each pressure in excess of the limit specified in 3.8.6.2(b).

c. The engaged coupling and nozzle shall be subjected to an inlet fluid pressure of 120±2 psig and a fluid flow of 100 -5/+0 to 600 -0/+30 gal/min in increments of 100 gal/min. The

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nozzle shall be disengaged at every increment. The spillage per disengagement shall not be greater than the limit specified in 3.8.6.2(c).

d. The coupling shall be subjected to an inlet fluid pressure of 10±1 psig providing unrestricted fuel flow. The nozzle shall be engaged and disengaged a minimum of five times. Spillage per engagement or disengagement shall not be greater than the limit specified in 3.8.6.2(d).

4.3.8.7 Pressure drop. The test shall be conducted in accordance with SAE ARP868, Single or Double Piezometer Tube Method, or equivalent, as approved by the qualifying activity with the coupling engaged with an MA-2 nozzle (Eaton, part number 60318-2B), all regulators inactive, and a pipe inner diameter of 3.0±0.1 inches. The pressure drop shall be measured at 50-gal/min increments over the flow-rate range of 0 to 600 -0/+30 gal/min (for report only). At 600 -0/+30 gal/min, the pressure drop shall not be greater than the value in section 3.8.7.

4.3.8.8 Extreme temperature. The coupling shall be subjected to tests in accordance with TABLE III.

4.3.8.8.1 Follow-on tests. Immediately after the test in 4.3.8.8, the coupling shall pass the functional tests of 4.3.8.15 to 4.3.8.15.6 and the leakage tests of 4.3.8.6.2 (subsequent).

4.3.8.9 Contaminated fuel. The test fluid shall contain the types and concentration of contaminants specified in TABLE IV. The engaged nozzle and coupling shall be subjected to a flow rate of at least 50 gal/min and operated for 1,000 cycles (a cycle shall consist of engaging for 30 seconds and disengaging). The flow-generating device shall ensure that the contaminants remain in uniform suspension in the fluid. Systems that use recirculated contaminants shall be designed such that contaminants are continuously recirculated back to the flow-generating device without falling out of suspension in the fluid. Upon completion, the coupling shall be flushed and drained.

4.3.8.9.1 Follow-on tests. Immediately after the tests in 4.3.8.9, the coupling shall pass the regulation-capability tests of 4.3.8.1.1, the leakage tests of 4.3.8.6.2 (subsequent), and the functional tests of 4.3.8.15 to 4.3.8.15.6.

4.3.8.10 Design limit loads. The coupling shall be subjected to the loads of 3.8.10. The tension loads shall be applied to the drogue attachment flange and the hose-attachment point, see figure 5. The compression loads shall be applied to the coupling through a fully-engaged MA-2 receiver aerial refueling probe nozzle, or dimensionally equivalent device, with the hose-end attachment fixed. The radial loads shall be applied on the coupling body 4.25±0.25 inches from the drogue attachment flange, see figure 10. No malfunction or deformation shall be evident upon completion of this test.

4.3.8.10.1 Follow-on tests. Immediately after the tests in 4.3.8.10, the coupling shall then be subjected to the functional tests of 4.3.8.15 to 4.3.8.15.6 and the leakage tests of 4.3.8.6.2 (subsequent).

4.3.8.11 Nozzle-coupling compatibility. The coupling shall be engaged with a nozzle so that the nozzle's sleeve latches are aligned with the coupling's locking roller windows in a

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manner which will produce the highest likelihood of binding or hang-up during engagement and disengagement. The force to engage the nozzle shall be measured. The nozzle shall then be disconnected and the disengagement force measured. No binding or hang-up shall occur. The engagement force shall be in accordance with 3.8.3.1. The disengagement force shall be within ±30 pounds of the requirement defined in 3.8.3.2. The coupling and the nozzle shall then be re-engaged. With the nozzle held fixed, the coupling shall be rotated by the outlet flange about its longitudinal axis and about the fixed nozzle through 360 degrees in both directions and the torque measured. No hang-ups or excess torque shall occur. This test shall be repeated using each nozzle on QPL-25161.

4.3.8.12 Altitude. The coupling, when engaged with an MA-2 nozzle (Eaton, part number 60318-2B), shall be subjected to a fluid flow of 400 gal/min at an inlet pressure of 80±2 psig and sea-level ambient pressure. While maintaining this condition, the ambient coupling pressure shall be reduced gradually (no slower than 1 psia per minute) to 4.17±0.4 psia. Sufficient data points (as agreed by the qualifying activity) shall be taken to plot a curve of regulated output versus ambient pressure. If the coupling is equipped with ambient-sensing ports, the ports shall then be closed (while the ambient pressure is maintained at 4.17±0.4 psia) and the test repeated. The regulated outlet pressure shall not decrease more than 15 psig, with and without the ambient-sensing ports closed. There shall be no external leakage of fuel from the ambient-sensing ports during the test.

4.3.8.13 Environmental requirements.

4.3.8.13.1 Salt fog. The coupling shall be installed in a test chamber in such a way that the outlet remains closed and the inlet is sealed against moisture. A salt solution of 5±1 percent sodium chloride in distilled water shall be atomized within the test chamber. At least 0.3 quart of salt solution shall be atomized per cubic foot of test chamber volume for 24 hours. Fallout rate of the salt fog shall be measured at least once every 24 hours, and shall be between 0.1 and 0.61 teaspoon/hour per 12.4 in^2 of collection area. Test chamber temperature and salt solution temperature prior to atomization shall be maintained at 95±5 °F. The coupling shall be exposed to salt fog for 48 hours, after which it shall be dried in ambient air as specified in 4.2.1 for 48 hours. The coupling shall then be flushed with warm water, dried, and wetted with test fluid. Corrosion shall be cause for rejection.

4.3.8.13.1.1 Follow-on tests. Immediately after the test in 4.3.8.13.1, the coupling shall pass the functional tests of 4.3.8.15 to 4.3.8.15.6 and the leakage tests of 4.3.8.6.2 (subsequent).

4.3.8.13.2 Sand and dust. The coupling shall be subjected to blown sand and dust in the manner specified in 4.3.8.13.2.1 to 4.3.8.13.2.2. At the conclusion of the test, the remaining loose sand and dust in or on the coupling shall be blown away.

4.3.8.13.2.1 Test conditions.

4.3.8.13.2.1.1 Sand composition. The sand particles used shall be of sub-angular structure with a mean roundness factor of 0.2 and a hardness factor of 7 (Mohs' scale). The sand shall consist of silica sand composed of at least 95 percent by weight silicon oxide with the following sized distribution:

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a. 1.0±0.5 percent shall be retained by a 20-mesh screen.

b. 1.7±0.5 percent shall be retained by a 30-mesh screen.

c. 14.8±1.0 percent shall be retained by a 40-mesh screen.

d. 37.0±1.0 percent shall be retained by a 50-mesh screen.

e. 28.6±1.0 percent shall be retained by a 70-mesh screen.

f. 12.7±1.0 percent shall be retained by a 100-mesh screen.

g. The remainder shall pass through a 100-mesh screen.

4.3.8.13.2.1.2 Dust composition. The dust used shall consist of silica flour or red China clay with the following size distribution:

a. 100 percent shall pass through a 100-mesh screen.

b. 98±2 percent shall pass through a 140-mesh screen.

c. 90±2 percent shall pass through a 200-mesh screen.

d. 75±2 percent shall pass through a 325-mesh screen.

4.3.8.13.2.1.3 Air velocity. For the sand test, the air velocity shall be maintained between 3,450 and 5,700 feet per minute (ft/min). For the dust test, the air velocity shall be maintained between 300 and 500 ft/min.

4.3.8.13.2.1.4 Sand and dust concentrations. Sand concentration shall be maintained at 0.961±0.218 grains per cubic foot (gr/ft^3). Dust concentration shall be maintained at 4.632 gr/ft^3.

4.3.8.13.2.2 Test procedure. The coupling's inlet shall be sealed, and the coupling shall be mounted in such a way as to allow sand and dust to be blown upon the coupling from different directions. Each test shall be run two times: once with the coupling's outlet pointed directly into the flow, and once with the coupling's axis aligned perpendicular to the flow. The coupling shall be subjected to blowing sand for 90 minutes from each direction. The coupling shall then be subjected to blowing dust for 6 hours at room temperature and 6 hours at an ambient air temperature of 185±10 °F. The coupling's temperature shall be allowed to stabilize at the ambient air temperature prior to each test run.

4.3.8.13.2.3 Follow-on tests. Immediately after the test in 4.3.8.13.2, the coupling shall pass the functional tests of 4.3.8.15 to 4.3.8.15.6 and the leakage tests of 4.3.8.6.2 (subsequent).

4.3.8.13.3 Vibration. The vibration test shall be performed in accordance with MIL-STD-810, Test Method 514.7, Procedure IV, and figure 7. This test shall be performed with the coupling in a dry condition and mounted to the test fixture in a similar manner as shown in figure 11. For the purpose of this test, 30 minutes of vibration in each axis shall be performed with a KC-130, low-speed, drogue basket, NSN 1680-00-068-1356 (supplied by the qualifying

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activity), and another 30 minutes vibration in each axis with a KC-130, high-speed, drogue basket, NSN 1670-01-422-3223 (supplied by the qualifying activity).

4.3.8.13.3.1 Follow-on tests. Immediately after the test in 4.3.8.13.3, the coupling shall pass the functional tests of 4.3.8.15 to 4.3.8.15.6 and the leakage tests of 4.3.8.6.2 (subsequent).

4.3.8.13.4 Impact and endurance.

4.3.8.13.4.1 Impact. The coupling shall be installed upon the fixture shown in figure 9, or other setup approved by the qualifying activity, and subjected to the impact conditions described in TABLE V. Unless otherwise specified, an internal fluid pressure of 10±2 psig shall be maintained in the coupling throughout this test.

4.3.8.13.4.2 Endurance. The coupling shall be installed upon the fixture shown in figure 9, or other setup approved by the qualifying activity, and subjected to the endurance conditions described in TABLE V. Unless otherwise specified, an internal fluid pressure of 10±2 psig shall be maintained in the coupling throughout this test.

4.3.8.13.4.2.1 Follow-on tests. Immediately after the test in 4.3.8.13.4.2, the coupling shall pass the functional tests of 4.3.8.15 to 4.3.8.15.6 and the leakage tests of 4.3.8.6.2 (subsequent).

4.3.8.14 Disassembly and inspection. Upon completion of testing, the coupling shall be disassembled and all parts inspected. Parts which are distorted, weakened, or are abnormally worn shall constitute failure.

4.3.8.15 Functional.

4.3.8.15.1 Deflection. The coupling shall be deflected ±22.5 degrees from the coupling centerline about the swivel-ball joint. The torque required to deflect the coupling about the swivel-ball joint shall not be less than 98 in-lbs or greater than 216 in-lbs.

4.3.8.15.2 Disengaged leakage. The coupling shall be checked at static pressures of 5±0.5 psig and 60±2 psig for 1 minute at each pressure. The coupling shall meet the requirements of 3.8.15.2.

4.3.8.15.3 Engaged leakage. The coupling, with a nozzle engaged, shall be pressurized to 5±0.5 psig and 60±2 psig for 1-minute periods each. The coupling shall meet the requirements of 3.8.15.3.

4.3.8.15.4 Pressure relief. If the coupling incorporates a pressure-relief valve, a pressure of 160±2 psig shall be applied to the nozzle while maintaining 60±2 psig at the coupling inlet. Coupling inlet pressure variations shall be monitored, and a record of its variation with time shall be recorded. Pressure shall be relieved in accordance with 3.8.15.4.

4.3.8.15.5 Regulators. The coupling shall be engaged with a nozzle, using an apparatus similar to that shown in figure 8, and one of the regulators (regulator No. 1) disabled. The coupling shall then be pressurized to 80±2 psig and a flow rate of 10±0.5 cc/min established. The

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coupling inlet and outlet pressures shall be monitored to confirm that the active regulator has regulated the outlet pressure to 60 psig or less. The flow rate shall then be reduced to 10 drops per minute and the outlet pressure confirmed to not increase more than 8 psig. The tests above shall be performed again with the disabled regulator (regulator No. 1) re-enabled and the other regulator (regulator No. 2) disabled.

4.3.8.15.6 Engagement and disengagement. The coupling latching mechanism shall be adjusted to produce a nominal disengaging force of 420 pounds at 0 psig fuel pressure, unless otherwise specified in the contract. A testing and production tolerance of ±20 pounds will be allowed for the disengaging force. The following tests shall then be conducted:

a. A fuel pressure of 50±2 psig shall be applied to the coupling outlet and the latching mechanism shall be checked. The disengagement force shall increase 200±45 pounds from the nominal setting at zero fuel pressure as specified in 3.8.3.2.1.

b. Using a test stand similar to that shown in figure 9, the test coupling shall be filled with fuel and pressurized to 10±1 psig. The engaging force shall not be greater than 155 pounds as specified in 3.8.3.1.

c. With the nozzle engaged and while maintaining a 10±1 psig fuel pressure in the coupling, the nozzle shall be disengaged and engaged at least five times (nozzle does not need to be held at 10±1 psig fuel pressure while disengaged from coupling). No evidence of binding, chattering, or excessive forces shall occur as specified in 3.8.11.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.2 Marking of shipments. Interior and exterior containers shall be marked in accordance with MIL-STD-129.

5.2.1 Instructions. The following instructions shall be marked on the unit, intermediate packages, and shipping container:

IF IN STORAGE FOR 5 YEARS OR LONGER, THE COUPLING SHALL BE TESTED AND INSPECTED BEFORE USE.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use. The aerial refueling coupling covered by this specification is intended for use by tanker aircraft for refueling receiver aircraft by means of the probe-drogue method of aerial refueling. The coupling, which is extended on a hose from the tanker, is engaged by the nozzle of the receiver aircraft, thereby allowing the transfer of fuel. The coupling may also be used for defueling the tanker aircraft on the ground with the use of a special item of test equipment. The coupling provides a pressure regulated fuel supply.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Locations of markings (see 5.2).

c. The disengaging force required, if other than 420 pounds (see 3.8.3.2).

d. Packaging requirements (see 5.1).

e. Lot inspections/tests (suggested). One coupling selected by the inspector from each lot of 100 or fraction thereof that has passed the individual tests shall be subjected to the tests in TABLE VIII.

f. Delivery of the coupling maintenance manual, to include a list of single-use parts.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 57267 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Air Force Life Cycle Management Center (AFLCMC), EZFA, 2145 Monahan Way, Bldg 28, Wright-Patterson Air Force Base, OH 45433-7017. An online listing of products qualified to this specification may be found in the Qualified Products Database) at https://assist.dla.mil.

6.4 Definitions.

6.4.1 Abnormal wear. Any unusual, excessive, or premature wear that would potentially cause the unit to function improperly.

6.4.2 External leakage. Leakage sufficient to form a drop after 6 minutes.

6.4.3 Hydrostatic/static pressure. The pressure of a fluid with no dynamic effects, usually measured with a port perpendicular to the fluid flow.

6.4.4 Qualifying activity. The engineering home office of the specification’s primary custodian.

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6.4.5 Routine maintenance. Simple, small-scale activities associated with regular and general upkeep of the unit.

6.4.6 Shroud. The aerodynamic fairing that also serves to protect the coupling body.

6.4.7 Swivel-ball joint. The component of the coupling assembly that is the interface between the hose-end fitting and the coupling body that allows for freedom of radial deflection in all directions, and 360 degrees of axial rotation relative to the centerline axis of the coupling (shown in figure 5).

6.5 Government-furnished and Government-loaned property. The contracting officer should arrange to loan the property listed in 3.2.

6.6 Subject term (key word) listing.

Air-to-air
Coupling
Drogue
Fuel
Hose reel
Probe

6.7 International standardization agreement implementation. This specification implements AIR STD 25/13, In-Flight (Aerial) Refueling Equipment, Dimensions and Functional Characteristics and STANAG 3447, Air-to-Air (Aerial) Refueling Equipment: Probe-Drogue Interface Characteristics. When amendment, revision, or cancellation of this specification is proposed, the preparing activity must coordinate the action with the U.S. National Point of Contact for the international standardization agreement, as identified in the ASSIST database at https://assist.dla.mil.

6.8 Changes from previous issue. This is a new specification, therefore, marginal notations are not used to identify changes.

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TABLE I. Qualification sequence No. 1.

Verification	Paragraph
Assembly	4.3.3
Corrosion-resistant materials → Dissimilar metals	4.3.4.2 to 4.3.4.5
O-rings, packings, and seals → Prohibited locking	4.3.7.1 to 4.3.7.2.2.3
Finishes	4.3.7.4 to 4.3.7.4.1
Maintainability → Interchangeability	4.3.7.6 to 4.3.7.7
Lubrication → Incorrect assembly	4.3.7.9 to 4.3.7.12
Pressure-sensing port	4.3.8.1.5
Weight	4.3.7.3
Dimensions	4.3.7.8
Functional (Initial)	4.3.8.15 to 4.3.8.15.6
Leakage and spillage (Initial)	4.3.8.6.1
Proof pressure	4.3.8.4 to 4.3.8.4.2.1
*Leakage and spillage (Initial)	4.3.8.6.1
*Functional (Initial)	4.3.8.15 to 4.3.8.15.6
Pressure drop	4.3.8.7
Fluid resistance and Extreme temperature	4.3.4.1 & 4.3.8.8 to 4.3.8.8.1
*Leakage and spillage (Subsequent)	4.3.8.6.2
*Functional (Subsequent)	4.3.8.15 to 4.3.8.15.6
Deicing fluid resistance	4.3.4.1.1 to 4.3.4.1.1.1
*Leakage and spillage (Subsequent)	4.3.8.6.2
*Functional (Subsequent)	4.3.8.15 to 4.3.8.15.6
Regulation	4.3.8.1 to 4.3.8.1.7 (except 4.3.8.1.5)
Contaminated fuel	4.3.8.9 to 4.3.8.9.1
*Regulation capability	4.3.8.1.1
*Leakage and spillage (Subsequent)	4.3.8.6.2
*Functional (Subsequent)	4.3.8.15 to 4.3.8.15.6
Alignment	4.3.8.2
Latching mechanism	4.3.8.3 to 4.3.8.3.2.4
*Leakage and spillage (Subsequent)	4.3.8.6.2
*Functional (Subsequent)	4.3.8.15 to 4.3.8.15.6
Design limit loads	4.3.8.10 to 4.3.8.10.1
*Leakage and spillage (Subsequent)	4.3.8.6.2
*Functional (Subsequent)	4.3.8.15 to 4.3.8.15.6
Ultimate (burst) pressure	4.3.8.5 to 4.3.8.5.2.1
*Leakage and spillage (Subsequent)	4.3.8.6.2
Electrical protection	4.3.4.6 to 4.3.4.6.5
Disassembly and inspection	4.3.8.14
Reliability	4.3.7.5

* Indicates a required follow-on test.

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TABLE II. Qualification sequence No. 2.

Verification	Paragraph	1 Unit	2 Units
Assembly	4.3.3	A	A & B
Corrosion-resistant materials → Dissimilar metals	4.3.4.2 to 4.3.4.5	A	A & B
O-rings, packings, and seals → Prohibited locking	4.3.7.1 to 4.3.7.2.2.3	A	A & B
Finishes	4.3.7.4 to 4.3.7.4.1	A	A & B
Maintainability → Interchangeability	4.3.7.6 to 4.3.7.7	A	A & B
Lubrication → Incorrect assembly	4.3.7.9 to 4.3.7.12	A	A & B
Weight	4.3.7.3	A	A & B
Dimensions	4.3.7.8	A	A & B
Functional (Initial)	4.3.8.15 to 4.3.8.15.6	A	A & B
Leakage and spillage (Initial)	4.3.8.6.1	A	A & B
Nozzle-coupling compatibility	4.3.8.11	A	A & B
Altitude	4.3.8.12	A	A & B
Salt fog	4.3.8.13.1 to 4.3.8.13.1.1	A	A
*Leakage and spillage (Subsequent)	4.3.8.6.2	*	*
*Functional (Subsequent)	4.3.8.15 to 4.3.8.15.6	*	*
Sand and dust	4.3.8.13.2 to 4.3.8.13.2.3	A	A
*Leakage and spillage (Subsequent)	4.3.8.6.2	*	*
*Functional (Subsequent)	4.3.8.15 to 4.3.8.15.6	*	*
Vibration	4.3.8.13.3 to 4.3.8.13.3.1	A	B
*Leakage and spillage (Subsequent)	4.3.8.6.2	*	*
*Functional (Subsequent)	4.3.8.15 to 4.3.8.15.6	*	*
Impact and endurance	4.3.8.13.4 to 4.3.8.13.4.2.1	A	B
*Leakage and spillage (Subsequent)	4.3.8.6.2	*	*
*Functional (Subsequent)	4.3.8.15 to 4.3.8.15.6	*	*
Ultimate (burst) pressure	4.3.8.5 to 4.3.8.5.2.1	A	A & B
*Leakage and spillage (Subsequent)	4.3.8.6.2	*	*
Electrical protection	4.3.4.6 to 4.3.4.6.5	A	A & B
Disassembly and inspection	4.3.8.14	A	A & B
Reliability	4.3.7.5	A	A & B

* Indicates a required follow-on test.

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TABLE III. Fluid resistance and extreme temperature.

Test Period and Duration[1]	**Test Fluid and Temperature**	**Operation During Period**	**Operation and Tests Immediately After Period**
Phase I - Soak[2] 96 hours (coupling is disengaged during the first 72 hours, and engaged with nozzle the last 24 hours)	ASTM D5900, IRM 903 158±2 °F	First 72 hours: Engage then disengage nozzle with coupling every 6 hours. Last 24 hours: Disengage then engage nozzle with coupling every 6 hours. At no time during the test shall the force to engage nozzle be greater than 155 pounds.	Perform functional tests as specified in 4.3.8.15 to 4.3.8.15.6 using ASTM D5900, IRM 903.
Phase I - Dry (coupling only, drained and blown dry) 24 hours	ASTM D5900, IRM 901 Circulating air at 158±2 °F	None	Perform functional tests as specified in 4.3.8.15 to 4.3.8.15.6 using ASTM D5900, IRM 901.
Phase II - Soak 18 hours (coupling and nozzle disengaged)	ASTM D5900, IRM 903 158±2 °F	Engage then disengage nozzle with coupling every 6 hours. At no time during the test shall the force to engage nozzle be greater than 155 pounds.	Perform functional tests as specified in 4.3.8.15 to 4.3.8.15.6 using ASTM D5900, IRM 903.
Phase II - Dry (coupling only, drained and blown dry) 30 hours	ASTM D5900, IRM 901 Circulating air at 158±2 °F	None	Perform functional tests as specified in 4.3.8.15 to 4.3.8.15.6 using ASTM D5900, IRM 901.

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TABLE III. Fluid resistance and extreme temperature – Continued.

Test Period and Duration[1]	Test Fluid and Temperature	Operation During Period	Operation and Tests Immediately After Period
Phase III – Soak (coupling and nozzle disengaged) 18 hours	ASTM D5900, IRM 901 -67±2 °F	None	Maintaining the ambient air and fluid temperature at -67±2 °F, the nozzle shall be engaged with the coupling and examined for external leakage while maintaining the pressure conditions of 4.3.8.15.2 for 15 minutes each. There shall be no external leakage. Then immediately with the ambient temperature and test fluid stabilized at –40±5 °F, the nozzle shall be engaged and disengaged from the coupling 50 times. During this test, a fluid pressure of 10 psig shall be maintained in the coupling and nozzle. Upon completion, there shall be no leakage after 15 minutes at 5 and 60 psig pressure each in the engaged and disengaged condition. The total leakage shall be recorded.

NOTES:
1. Each period shall follow immediately after the preceding one in the order noted.
2. During soak periods in the test fluid, the coupling shall be filled with the test fluid to ensure complete contact of all parts with the fluid as would be expected under service conditions.

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TABLE IV. Contaminant mixture.

Contaminant	**Particle Size (Microns)**	**Quantity (grams per 1000 liters)**
Iron oxide	0 - 5 5 - 10	19.00 1.00
Sharp silica sand	150 - 300 300 - 420	0.70 0.70
Prepared dirt conforming to AC Spark Plug Co. Part No. 1543637 (coarse Arizona road dust, or equivalent)	Mixture as follows: 0 - 5 (12 percent) 5 - 10 (12 percent) 10 - 20 (14 percent) 20 - 40 (23 percent) 40 - 80 (30 percent) 80 - 200 (9 percent)	5.30
Cotton linters	Staple below 7 US Dept. of Agriculture Grading Standards	0.07
Iron chips Aluminum chips	150 - 500 150 - 500	10.00 10.00
Crude naphthenic acid		0.03 percent by volume
Salt water solution (5±1 percent by weight NaC1)		0.01 percent entrained

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TABLE V. Impact and endurance test.

Condition[1]	Displacement of coupling centerline from nozzle centerline (inches)	Velocity at initial contact between nozzle nose and coupling poppet (ft/s)[8]	Angle of disconnect (degrees)	Initial, dry latch-force setting[7] (lb + 20)	Cycles[2]
Impact[3]	0	9.8	0	500	[4]100
	2	9.8	0	500	400
	4	10.4	0	500	500
Endurance[5]	0	6.9	15	500	[4]1000
	0	6.9	15	500	[6]250
	0	6.9	0	500	[4]2750

NOTES:

1. Thirty percent of the cycles at each condition shall be performed at -67±2 °F.
2. A cycle is defined as one engagement and one disengagement.
3. During the impact test, engagement shall be complete. If impact does not cause engagement, manual force shall be applied as required to complete engagement.
4. Fifty of the specified cycles shall be accomplished with zero fuel pressure and a dry nozzle.
5. Drop test shall not be required. Mechanical engagement shall be permissible, provided engaging velocity is not less than 5 ft/s just prior to seating the sleeve upon the master coupling seal.
6. Disengagement shall be accomplished with 50±2 psig fuel pressure applied to the reception coupling.
7. The term “latch-force setting” refers to the overall force required to disengage a probe from the coupling, also referred to as the “disengagement force.”
8. When using a drop-test method as shown in figure 9, the following drop distances apply:

   9.8 ft/s = 18 inches

   10.4 ft/s = 20 inches

   6.9 ft/s = 9 inches

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TABLE VI. Acceptance sequence.

Verification	Paragraph
Assembly	4.3.3
Corrosion-resistant materials → Dissimilar metals	4.3.4.2 to 4.3.4.5
O-rings, packings, and seals → Prohibited locking	4.3.7.1 to 4.3.7.2.2.3
Finishes	4.3.7.4 to 4.3.7.4.1
Maintainability → Interchangeability	4.3.7.6 to 4.3.7.7
Lubrication → Incorrect assembly	4.3.7.9 to 4.3.7.12
Weight	4.3.7.3
Dimensions	4.3.7.8
Functional	4.3.8.15 to 4.3.8.15.6
Proof pressure	4.3.8.4 to 4.3.8.4.2.1
Regulation capability[1]	4.3.8.1.1 (a. and b. at 80 psig inlet pressure only and flow rates 10 cc/min and 1, 50, 200, 400, and 600 gal/min)
Electrical protection	4.3.4.6 to 4.3.4.6.5

NOTE:

1. Pressure measurements for acceptance testing may be made at the latch port if sufficient data (as agreed by the qualifying activity) is collected during qualification testing to correlate latch-port pressure measurements to measurements taken 30 inches downstream of the coupling.

TABLE VII. Verification tolerances.

Flow Rate	Tolerance		Pressure	Tolerance
≤ 1 gal/min	±5%		< 20 psi	±10%
> 1 gal/min	±2%		≥ 20 psi	±2 psi

TABLE VIII. Lot sequence.

Verification	Paragraph
Leakage and spillage (initial)	4.3.8.6.1
Regulation	4.3.8.1 to 4.3.8.1.7 (except 4.3.8.1.5)
Design limit loads	4.3.8.10 to 4.3.8.10.1
*Leakage and spillage (subsequent)	4.3.8.6.2
*Functional (subsequent)	4.3.8.15 to 4.3.8.15.6
Ultimate (burst) pressure	4.3.8.5 to 4.3.8.5.2.1
*Leakage and spillage (subsequent)	4.3.8.6.2
Disassembly and inspection	4.3.8.14

* Indicates a required follow-on test.

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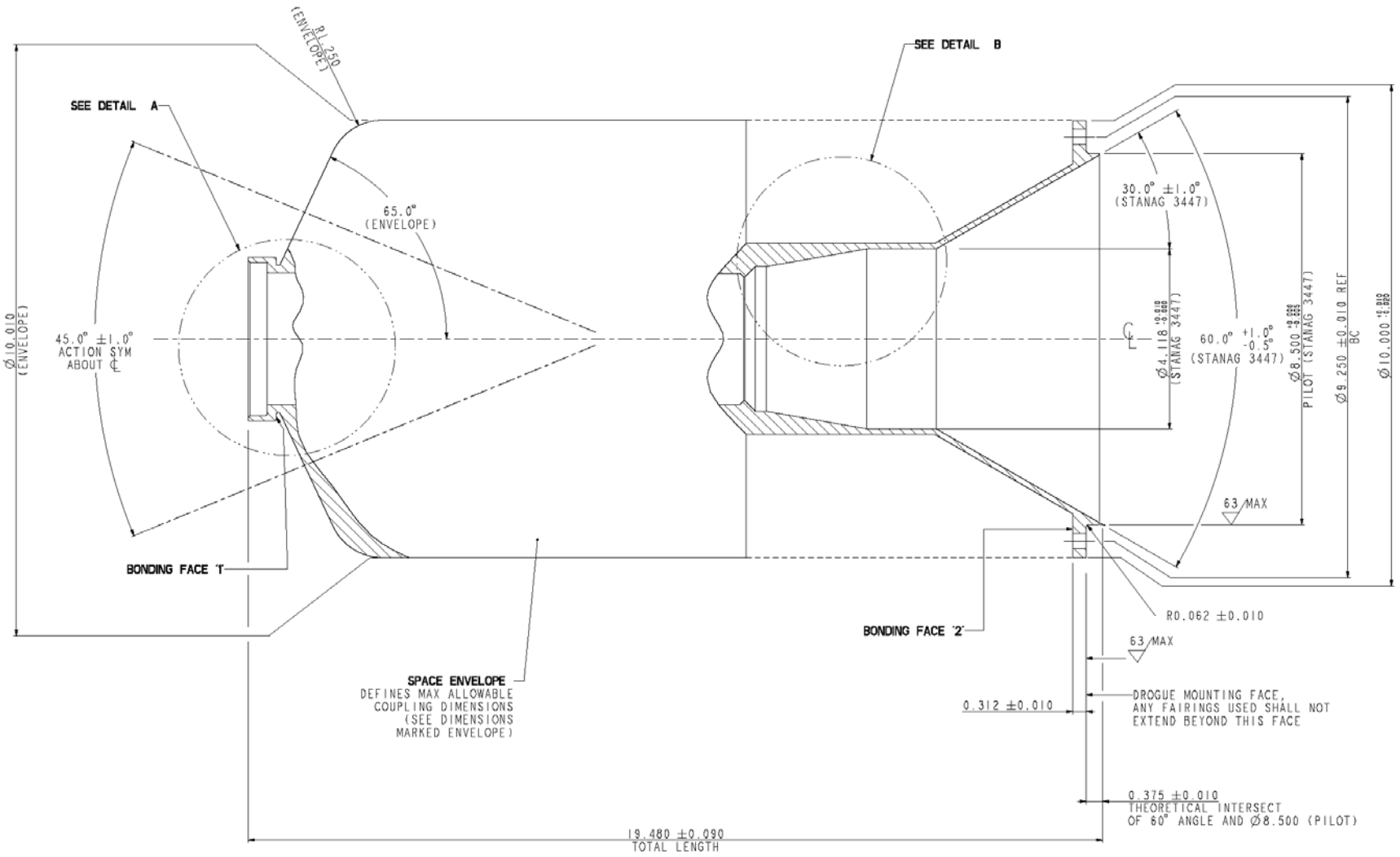


FIGURE 1. Coupling dimensions No. 1.

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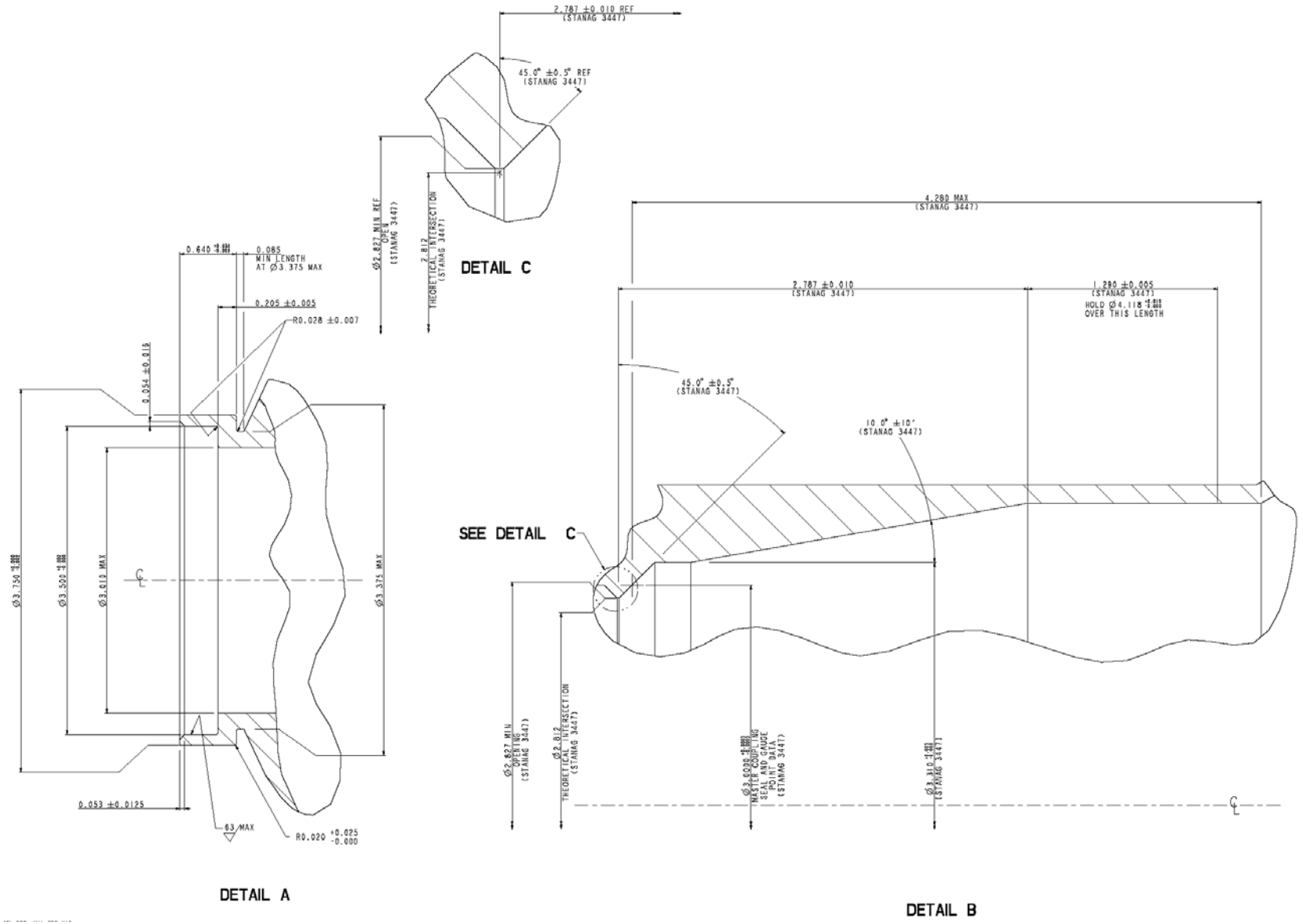
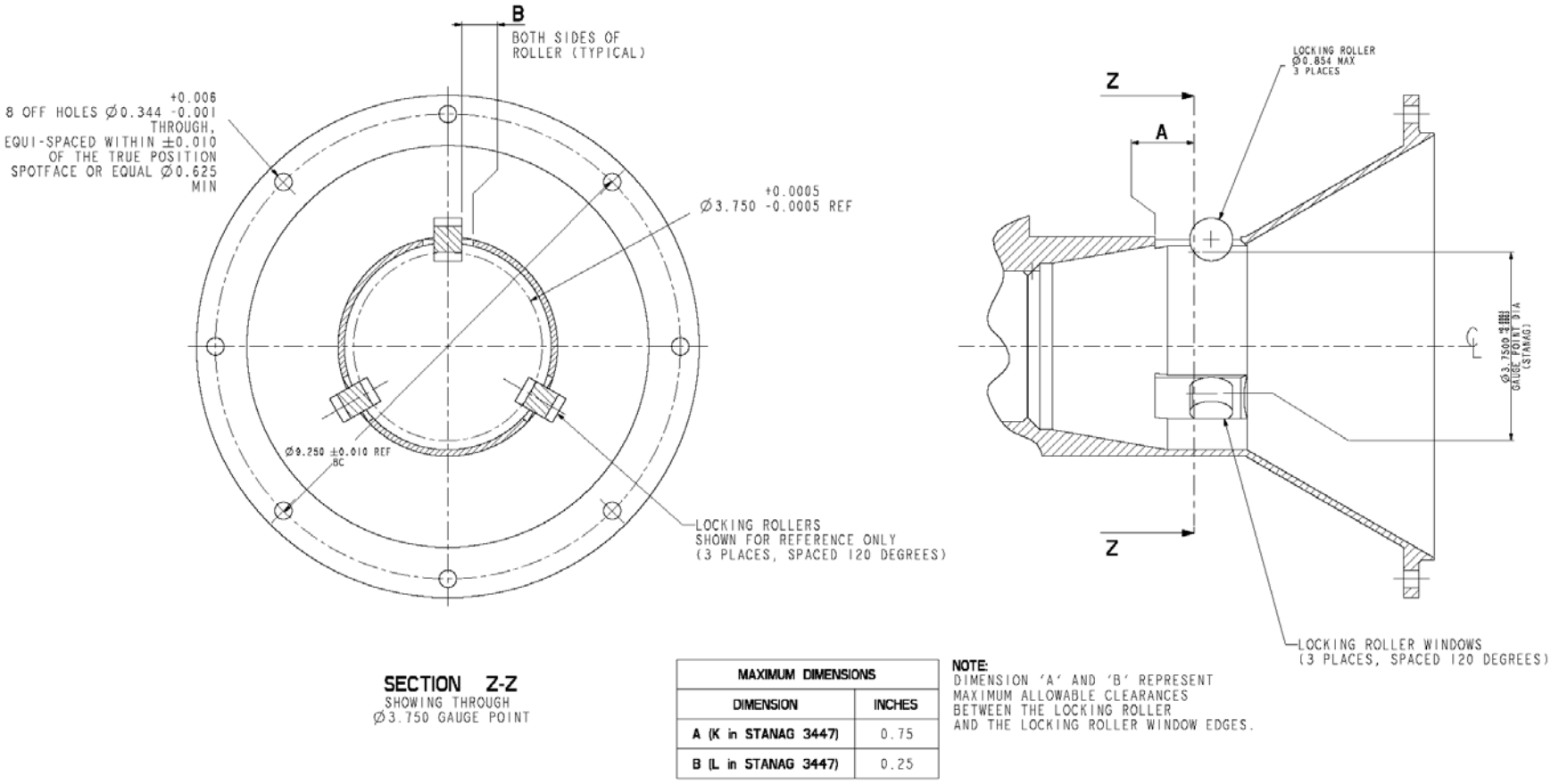


FIGURE 2. Coupling dimensions No. 2.

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MAXIMUM DIMENSIONS	
DIMENSION	INCHES
A (K in STANAG 3447)	0.75
B (L in STANAG 3447)	0.25

NOTE:
DIMENSION 'A' AND 'B' REPRESENT
MAXIMUM ALLOWABLE CLEARANCES
BETWEEN THE LOCKING ROLLER
AND THE LOCKING ROLLER WINDOW EDGES.

FIGURE 3. Coupling dimensions No. 3.

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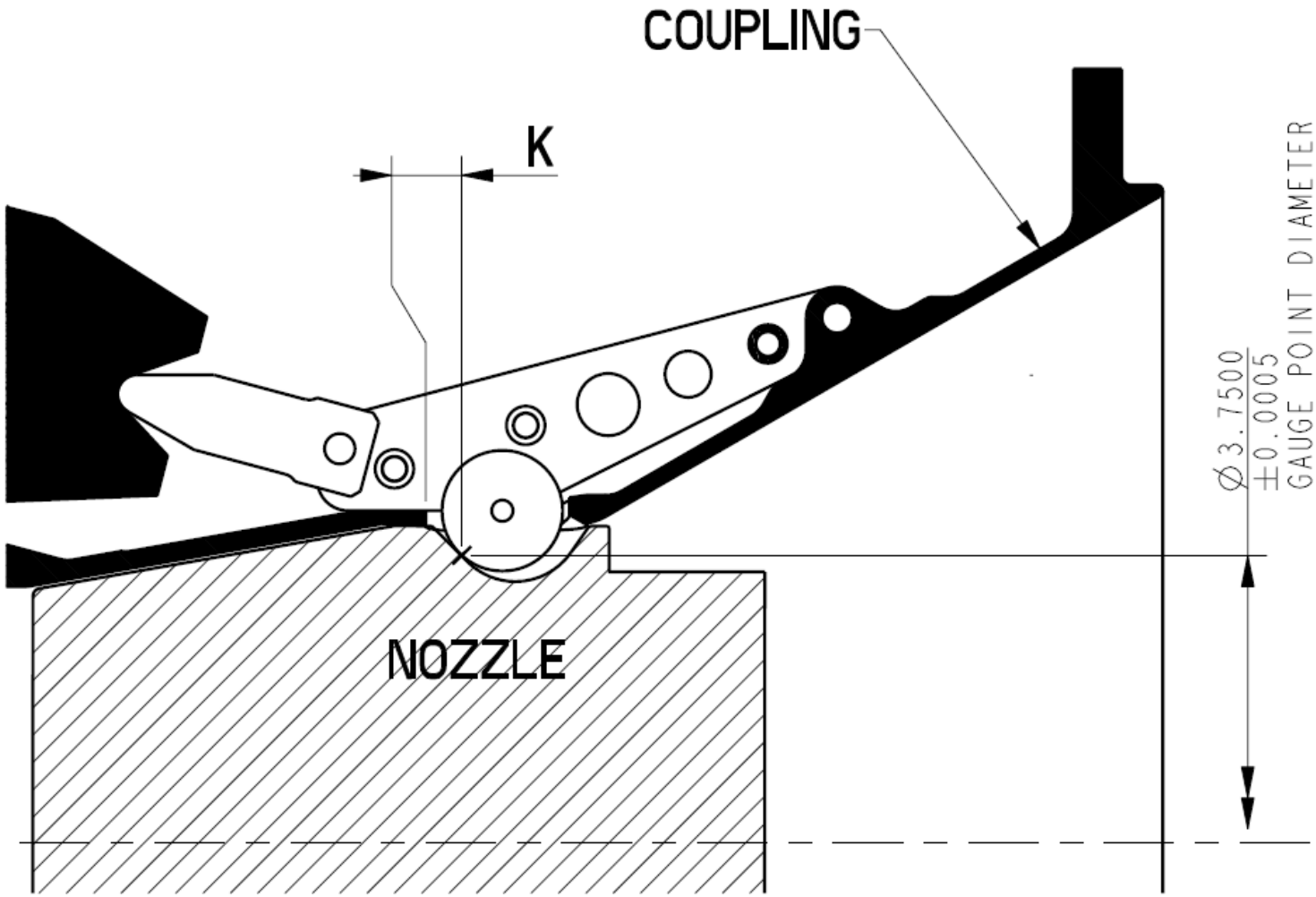


FIGURE 4. Gauge point dimension.

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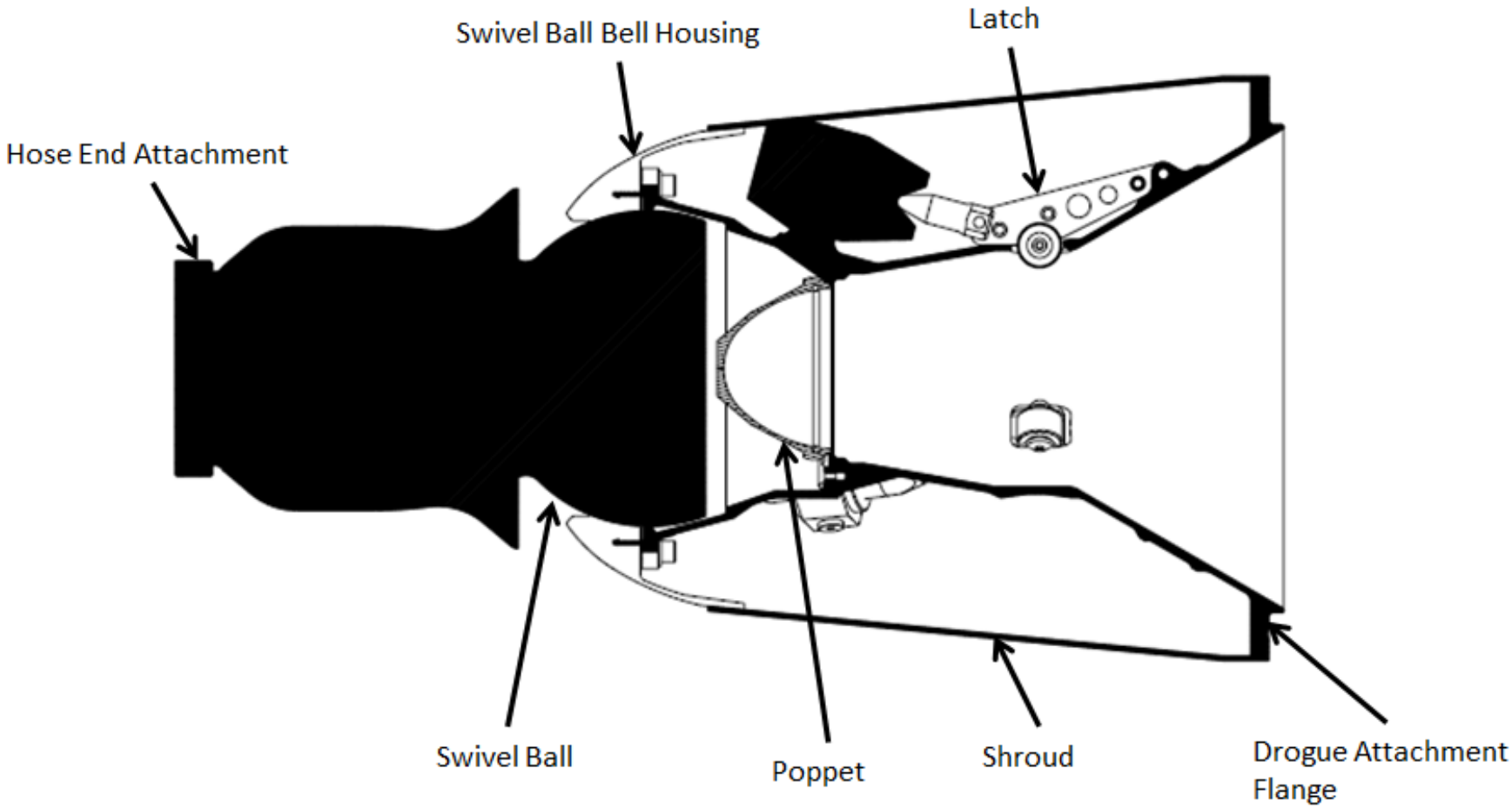


FIGURE 5. Coupling components.

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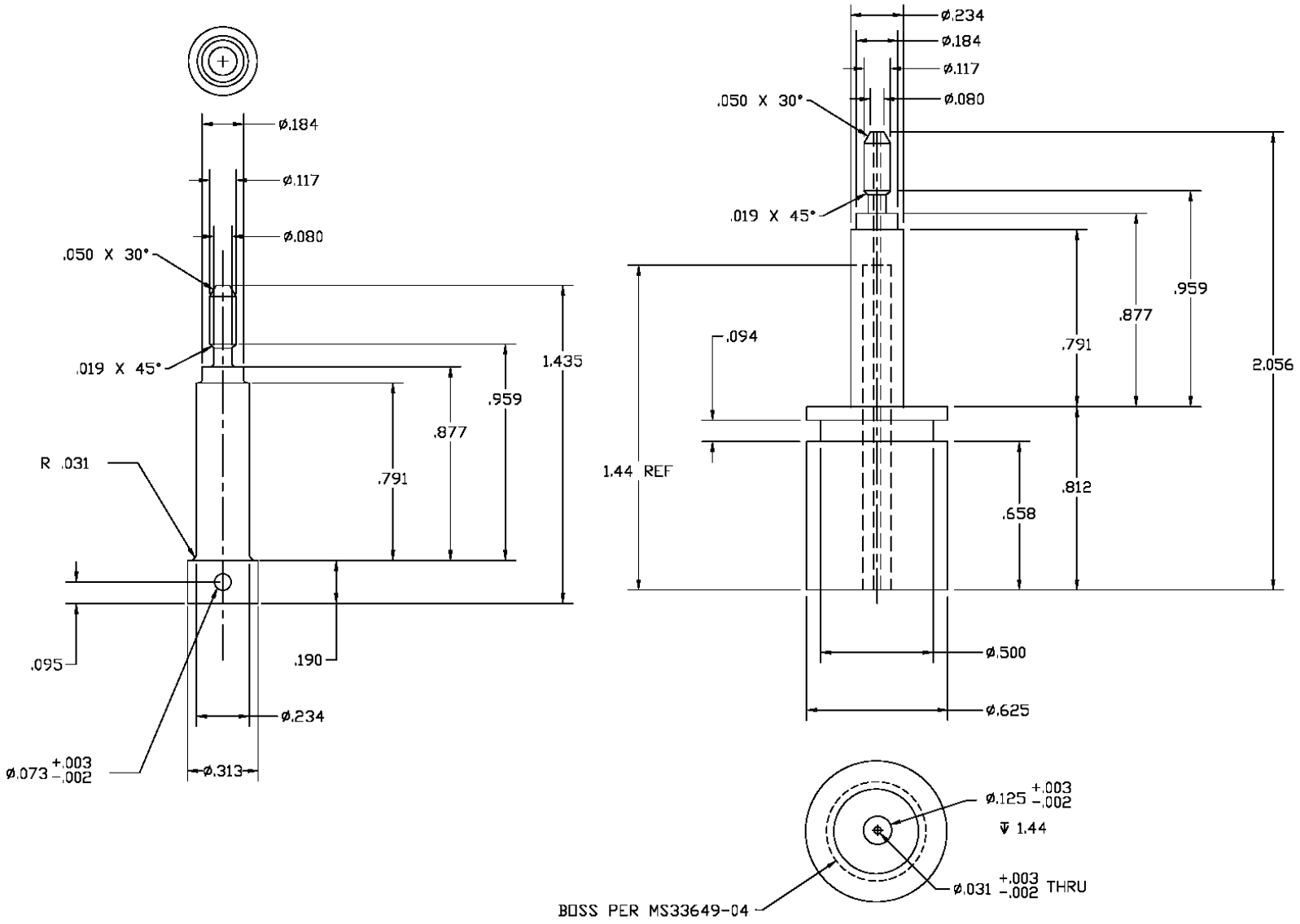


FIGURE 6. Ambient-sensing port plug dimensions.

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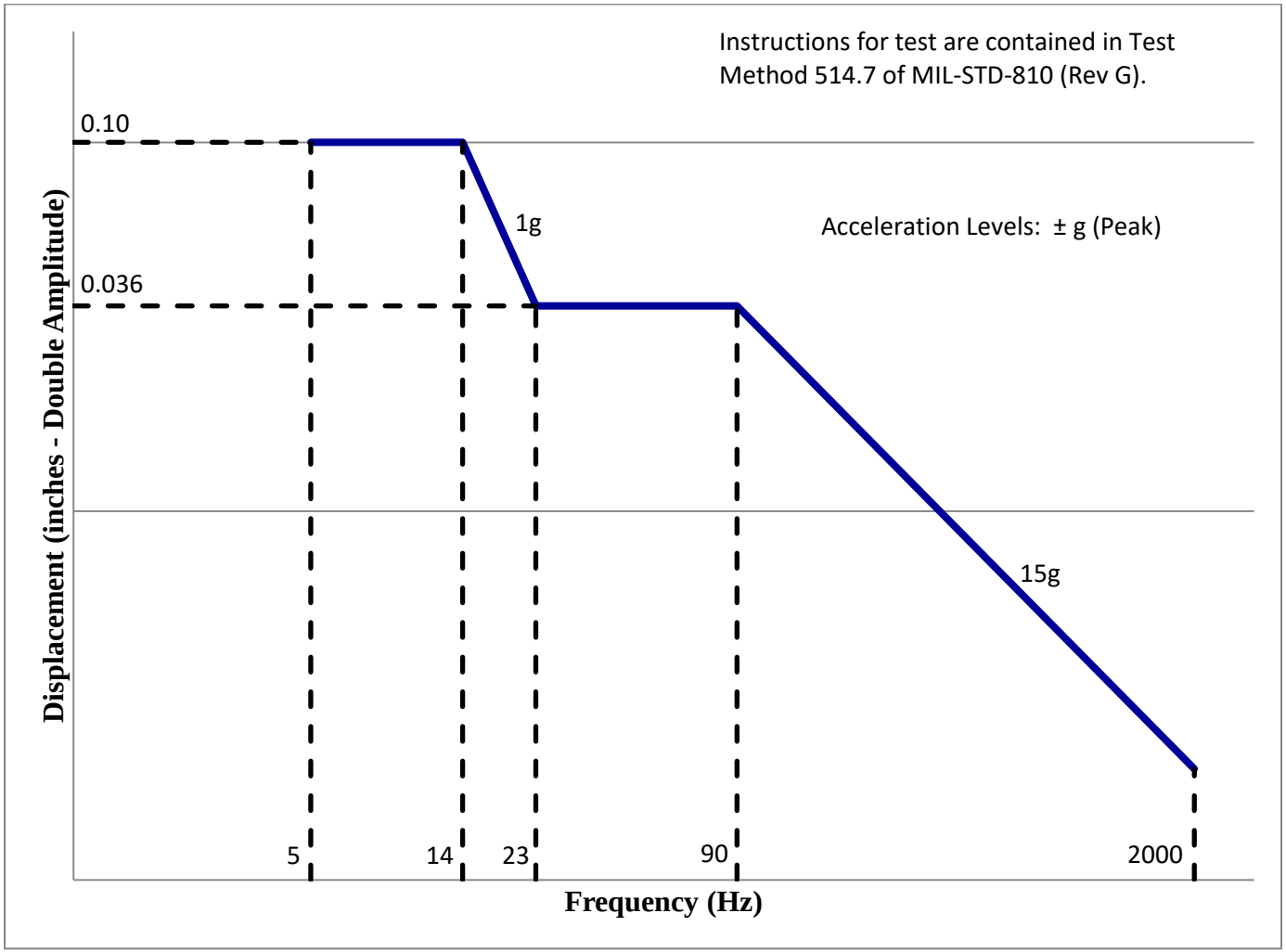
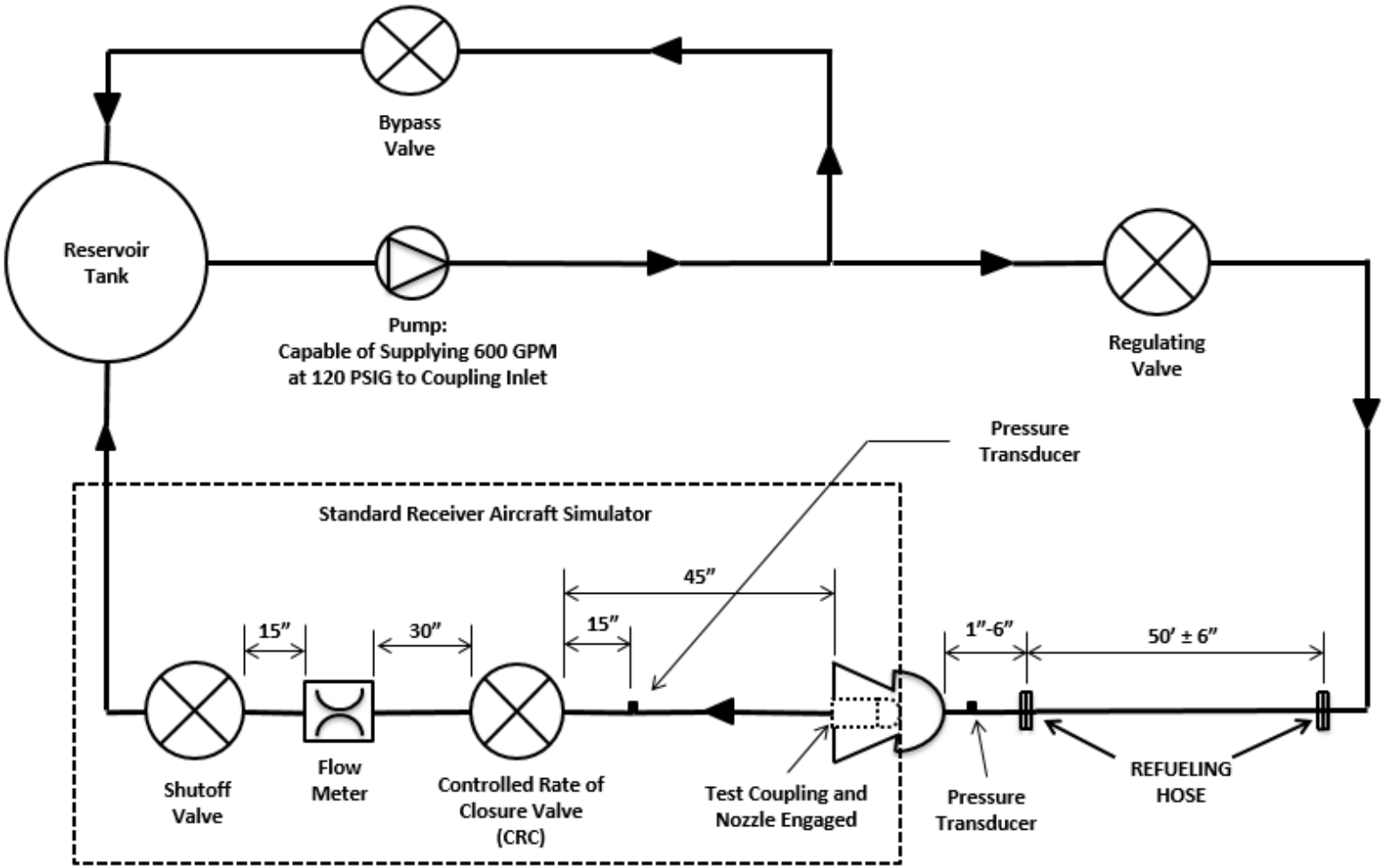


FIGURE 7. Vibration test curve.

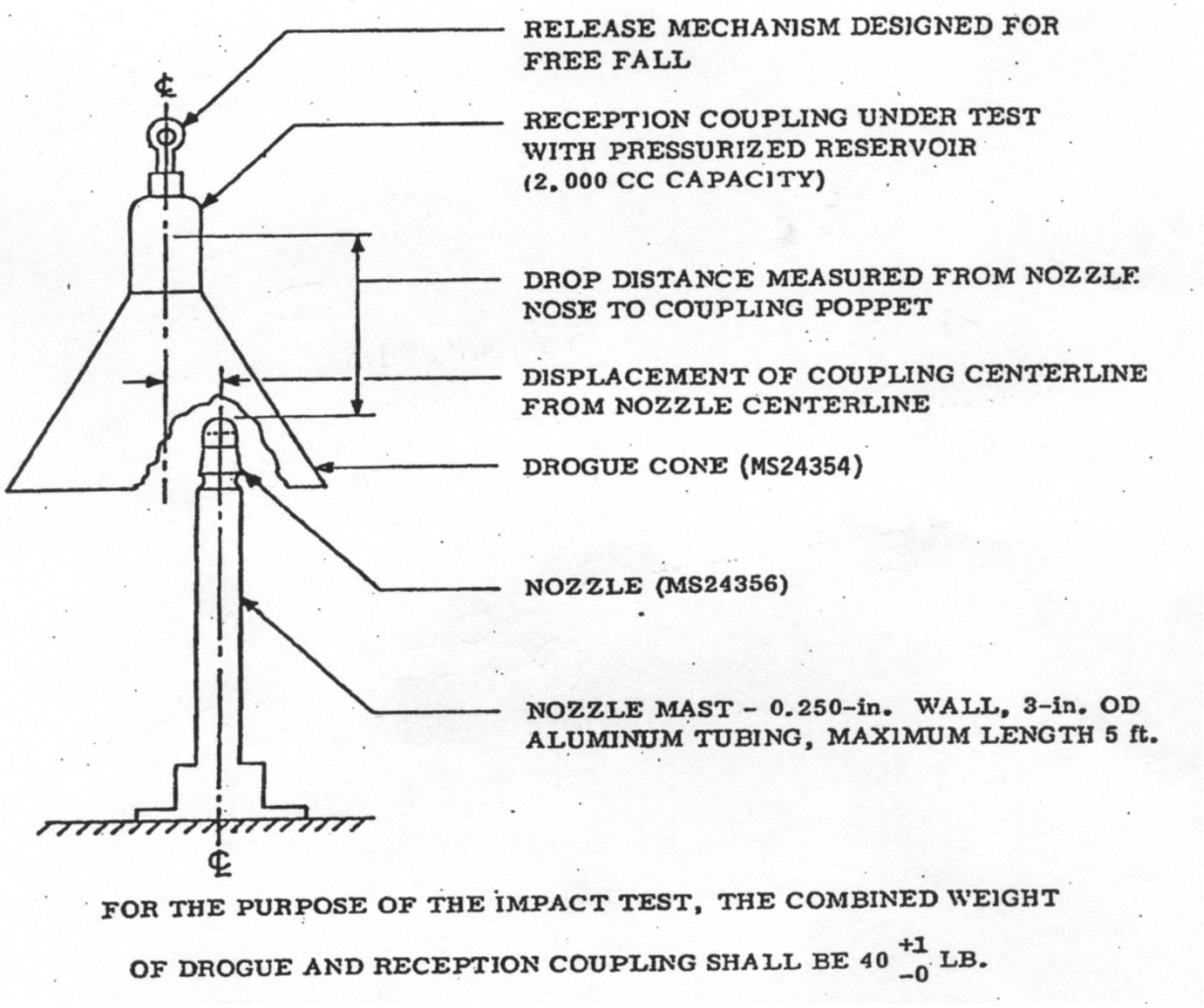
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NOTES:
1. All piping between the refueling hose and the CRC valve has an Inner Diameter (I.D.) of 3 inches ± 0.1 inch. All other piping has a minimum I.D. of 3 inches.
2. The hose is to be secured at both ends and lies straight along the ground as long as the test fixture is only 1-3 feet off the ground.
3. The refueling hose is not required for regulation tests (paragraph 4.3.6.1 and sub-paragraphs).
4. Unless otherwise specified distance dimensional tolerances shall be ± 0.5 inch.

FIGURE 8. Surge suppression and pressure regulation test setup.

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NOTE:
1. It is recommended that bungee cords be used to drop the coupling during the testing to simulate a more accurate representation of flight.

FIGURE 9. Impact and endurance test setup.

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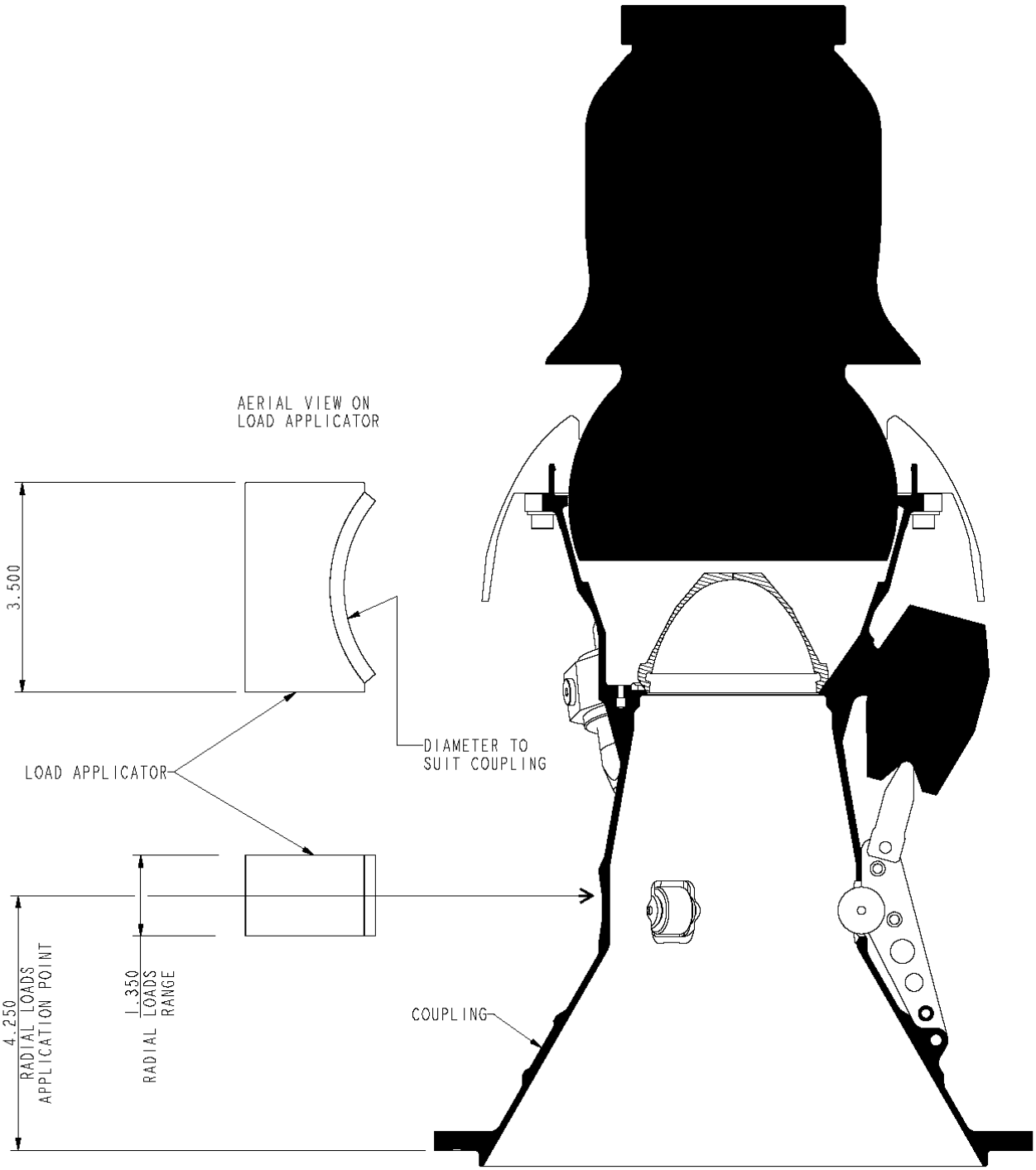


FIGURE 10. Load application points.

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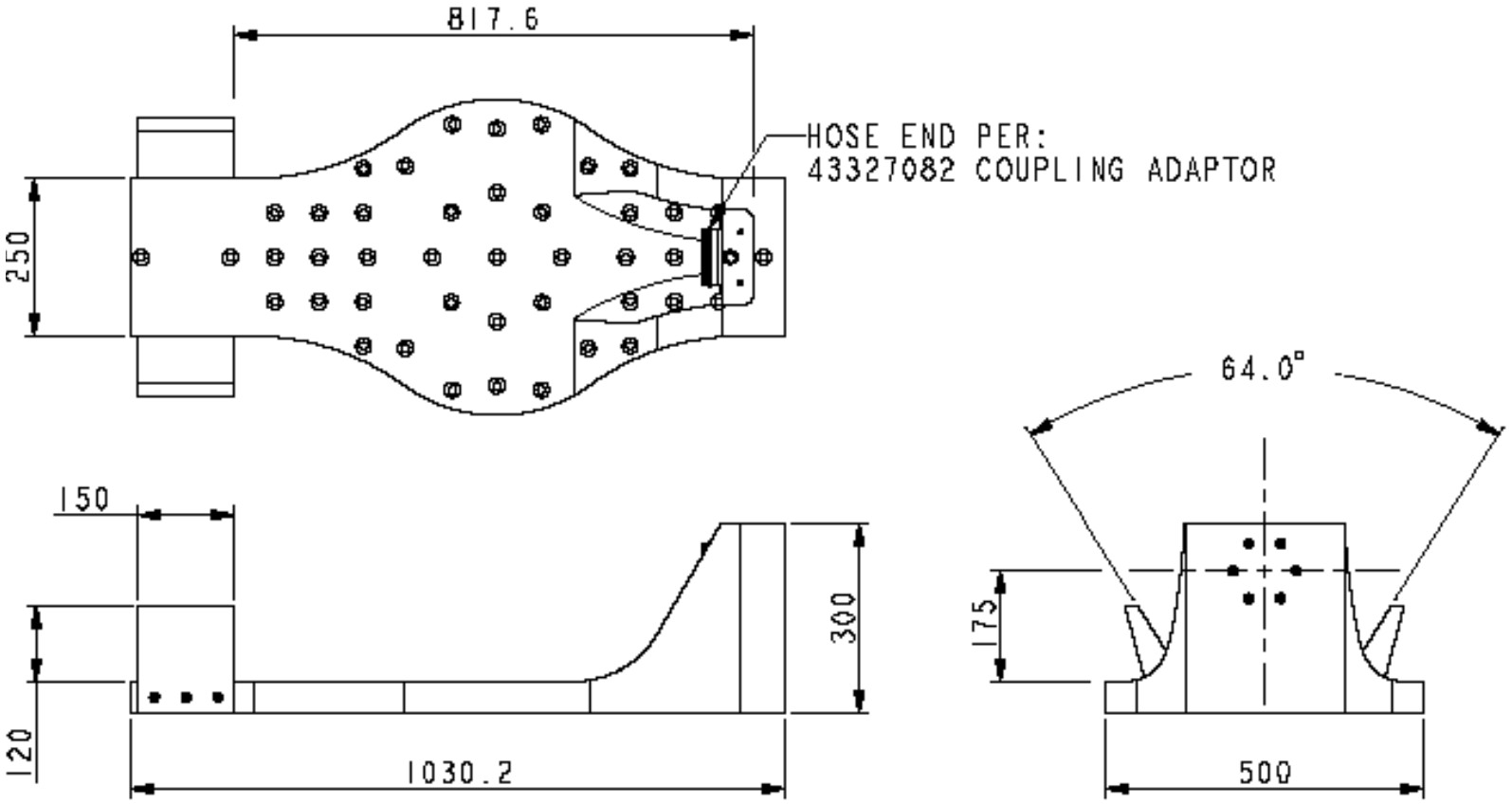


FIGURE 11. Vibration test setup.

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CONCLUDING MATERIAL

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Air Force – 11
DLA – GS

Preparing activity:
Air Force – 11

(Project 1680-2017-001)

Reviewers:
Army – AV
Air Force – 71

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